

MASTER

Sensory Modalities and Stress

Exploring How Sensory Ethnography Can Enrich Research On Psychophysiological Stress within Physical Workplace Environments

Laurens Janse, Annique J.

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Sensory Modalities and Stress: Exploring How Sensory Ethnography Can Enrich Research On Psychophysiological Stress within Physical Workplace Environments

Master Thesis (7Z45M0 + 1-IEIS-MSc)

Laurens Janse, A.J.
1244575

Supervisors:

prof.dr.ir. Yvonne de Kort
dr.ir. Rianne Appel-Meulenbroek
Sophie Schuller Msc

Graduation program: MSc Urban Systems & Real Estate (BE department) – Human Technology
Interaction (IE&IS department)
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SUMMARY

Workplace stress is a major concern in modern organizations, with significant implications for employee well-being, productivity, and mental health. Stress in workplace environments can stem from various factors, including workload, social interactions, and physical surroundings. While much research has focused on visual and auditory factors in workplace design, other sensory modalities such as smell and touch have been largely overlooked. This research therefore aims to investigate the role of smell and touch in shaping stress responses and to assess how subjective sensory experiences contribute to individual differences in stress perception. It explores how sensory ethnography—a research approach that emphasizes multisensory experiences—can enhance our understanding of psychophysiological stress in workplace settings. The primary research question guiding this study is:

"How can sensory ethnography incorporate lesser-researched senses to enhance research on workplace design and psychophysiological stress?"

The literature review revealed that visual and auditory stimuli in workplace design have been widely studied, while the role of olfaction and touch remains underexplored. Vision-related factors such as natural light, greenery, and clutter have been shown to influence stress levels, with natural elements generally reducing stress and visual clutter increasing it. Noise, particularly from human conversations, has been identified as a major workplace stressor, where acoustic privacy enhances well-being and the incorporation of nature sounds can be beneficial. The olfactory environment in workplaces has been largely ignored, though studies suggest that scents like lavender and citrus may reduce stress, albeit with highly individual responses. Touch, particularly the materiality and texture of office furniture and surfaces, has the potential to affect comfort, relaxation, and stress but has not been systematically investigated yet. Taste, although studied in food-related research, is not directly relevant to workplace design.

To explore how sensory ethnography can improve workplace stress research, several methodologies have been examined. Observations and participatory methods enable researchers to document how employees interact with sensory elements in real time. Interviews incorporating sensory elicitation encourage participants to describe their sensory experiences with prompts such as objects or images. Sensory diaries provide a structured yet flexible way for employees to log their daily sensory interactions, capturing recurring stressors or positive experiences. Participant-led photography allows individuals to document their sensory experiences visually, highlighting design elements that impact stress. Sensory mapping enables employees to chart their workplace perceptions, identifying areas that contribute to stress or relaxation. These methods offer new avenues for studying workplace stress through a multisensory lens.

From this literature review, two appropriate ethnographic (diaries and focus groups) methods have been chosen to pilot a mixed-methods approach that integrates both quantitative and qualitative methodologies. A 2x2 within-subject experimental design was used to manipulate sensory stimuli, specifically the presence or absence of smell and soft materiality. The study was conducted in a controlled workplace setting where participants experienced different environmental conditions. Psychophysiological stress was measured using heart rate variability (HRV). Self-reported stress was assessed through a validated questionnaire. In the diaries participants provided subjective evaluations of the sensory environment, and rating the appraisal of smell and softness. Additionally, semi-structured focus groups were conducted to collect qualitative data on participants' experiences and perceptions. The data was analysed using both statistical methods for quantitative findings and thematic analysis for qualitative insights.

Quantitative analysis showed that direct manipulation of smell and softness did not significantly affect physiological stress markers (RMSSD) or self-reported stress levels. However, subjective perception played a critical role—participants who rated the sensory environment more positively reported lower psychological stress levels tension and pleasure. This suggests that individual appraisal and personal associations with sensory inputs mediate stress responses rather than being determined by the stimuli themselves. Qualitative analysis provided further depth, revealing that smell often evoked strong emotional associations, with some participants finding certain scents relaxing while others experienced them as distracting or unpleasant. This highlights the subjective nature of olfactory perception in workplace stress. The presence of soft materials was generally perceived as comforting, but some participants felt that excessive softness reduced their alertness, suggesting that haptic experiences in workplace design must balance comfort and functionality. Many participants expressed a preference for minimal sensory stimulation, favouring neutral environments that allowed for higher focus and lower stress. These insights reinforce the need for adaptable and customizable workplace environments that cater to diverse sensory preferences.

By integrating sensory ethnographic methods with physiological and psychological stress measures, this study provided a richer, more nuanced perspective on how workplace environments shape employee stress levels. Sensory ethnographic methods like diaries and focus groups proved valuable for capturing nuanced experiences, though integrating them with physiological measures like HRV was challenging. The study highlights how sensory ethnography provides rich, contextual insights—particularly for less consciously experienced senses—and reveals how personal associations, homely elements, and even visual cues can shape sensory experiences. These results support using sensory ethnography to study subtle, subjective factors in workplace environments.

1. INTRODUCTION

1.1 Background and motivation

The physical workplace environment in relation to psych-physiological stress of the employees has gained increasing attention, in response to the rising prevalence of burnout and other mental health issues. As workplace stress contributes to more than half of all absenteeism (EU-OSHA., 2013), its reduction has become an important topic for employers, particularly with respect to improving employee well-being and performance (Gabriel & Aguinis, 2022). Psycho-physiological stress, which involves both psychological and physiological responses to stressors (Kropman et al., 2023), can have long-term physical health implications, including cardiovascular disease and inflammation (Pickering, 2001; Viljoen & Thomas (néé Negrao), 2021) as well as mental health implications such as anxiety, burn-out and depression (Ganster & Rosen, 2013).

The environment could play two different roles, it could be a stressor in itself when designed poorly, but it could also be a method of coping with (job) stress as it could have stress restoring effects (Kropman et al., 2023). Employees experience their environments through sensory perception, which plays a vital role in how the body and mind process physical surroundings (Gritzka et al., 2020; Spence, 2020a). While visual and auditory inputs have been commonly researched for their role in stress within the workplace (Bergefurt et al., 2021; Spence, 2020a), sensory modalities such as smell and touch are often underexplored (Spence, 2020a). Yet, these sensory inputs can also add to or mediate the relationship between the physical environment and stress, potentially influencing psychophysiological stress through different senses or even multisensory interactions (Comer et al., 2013; Lin et al., 2017). Despite this, sensory perception is frequently overlooked in research evaluating the impact of workplace environments on psychophysiological stress (Schuller, in press).

Many current studies rely on self-reported data, such as surveys and interviews, to evaluate the influence of workplace design on stress (Bergefurt et al., 2021). This approach, however, could present challenges, as participants may struggle to articulate the influence of lesser-studied sensory inputs like smell or touch (Soucek et al., 2023). Other studies that use physiological stress measurements in the workplace environments such as biomarkers, endocrine markers neurological indicators etc. can sometimes come with its limitations as well. Studying sensory perception in a naturalistic setting, such as an office environment, is challenging, particularly with the use of invasive or lab-based techniques (Schuller, in press). Most of those studies, for example studying biophilic design influences on stress, use a ‘plant on desk present or absent’ based method (Schuller, in press). But this ignores other biological mechanisms such as visual attention as well as psychological underlying mechanism of the stress reduction such as other senses, tacit experiences or personal associations (Schuller, in press). Traditional methods often do not shed light on this. As a result, key elements of the physical work environment’s influence on stress may be missing from these reports, leading to incomplete research

on workplace design and its effects on mental and physical well-being (Spence, 2020a). Mazzola et al. (2011) also emphasise the importance of qualitative methods in addition to quantitative measures, especially in work-related stress research. They argue that there can be any variety of stressors, which is why the provision of context and gaining a deeper understanding is key. This presents an opportunity to explore alternative research methodologies that are better suited to understanding the sensory dimension of workplace environments.

Ethnographic methods have long been used in workplace research to explore social interactions, cultural norms, and individual experiences (Gobo, 2011; Hampton et al., 2019). Sensory ethnography, a specialized form of ethnography, offers a promising approach to investigating the role of sensory perception in the workplace, focusing on how sensory modalities shape human experiences (Pink, 2015). By paying close attention to sensory data, such as touch, smell, and auditory inputs, sensory ethnography could provide richer insights into how these factors influence psycho-physiological stress in physical work environments (Holck & Villeseche, 2023; McGibbon et al., 2010; Pink, 2013).

The integration of sensory ethnographic techniques, such as observation, sensory diaries, interviews, and sensory mapping, into workplace research may allow for a more comprehensive exploration of how sensory stimuli interact with stress responses (McGibbon et al., 2010; Misharina et al., 2023; Pink, 2015). These techniques can capture subtle sensory experiences that traditional self-report measures often miss and can be placed in a more holistic environment compared to ‘plant present/absent’ methods, providing a more holistic understanding of the workplace’s impact on mental health (Mazzola et al., 2011; McGibbon et al., 2010). In particular, sensory ethnography allows for the observation and documentation of sensory modalities like touch and smell, which could affect stress responses (Spence, 2020a) but are often neglected in workplace studies (Pink, 2015; Spence, 2020a).

In conclusion, while psychophysiological stress in the workplace is a growing area of research, current studies fail to fully account for the role of sensory perception. Sensory ethnography offers a novel and potentially transformative approach to capturing the multisensory dimensions of the workplace and their influence on stress. Furthermore, it is widely used in practice and does not suffer from some of the difficulties of biomedical research methods (Pink, 2015). By expanding our understanding of how physical environments impact well-being through sensory perception, researchers can develop more effective workplace design strategies that foster healthier, more productive work environments (Lin et al., 2017; Pink, 2015; Spence, 2020a).

1.2 Research objective and questions

Considering the potential that sensory ethnographic methods could have for gaining more clarity on the effects of multi-sensory experiences in the workplace environment, the main research question is formulated as follows:

"How can sensory ethnography incorporate smell and touch and enhance research on workplace design and psycho-physiological stress?"

To be able to answer the main research question, multiple sub-questions are formulated:

1. What is the known influence of workplace design on psychophysiological stress and which senses are involved?
2. What are sensory ethnographic techniques that can measure touch and smell in the workplace and how are these techniques practiced?
3. Have these techniques been related to or used alongside of other measures of psycho-physiological stress such as biomarkers and self-reported stress?
4. Can sensory ethnographic methodologies add additional insights on workplace research on psychophysiological stress in relation to touch and smell?

1.3 Academic and practical relevance

This thesis contributes to the growing body of research on workplace-related psychophysiological stress by addressing a key gap in the literature: the underrepresentation of lesser-studied senses such as smell and touch in stress-related workplace studies. Through the application of sensory ethnography, it investigates a potentially innovative methodological approach that complements existing physiological and self-report measures by offering a richer, more contextualized understanding of multisensory experiences. By doing so, it might help advance the scientific discourse on how complex sensory interactions in real-world environments influence stress, and encourages a more holistic, embodied perspective in environmental psychology and workplace design research.

Concerning practical relevance, this thesis could offer insights for workplace designers, facility managers, and employers seeking to enhance employee well-being through evidence-based design. It suggests that workplace environments potentially should be designed with greater attention to sensory diversity, moving beyond visual and acoustic optimization to include haptic and olfactory dimensions. By applying sensory ethnographic techniques, organizations might uncover subtle stressors or soothing elements within their environments that are often overlooked. This may inform more nuanced design interventions that support both well-being and productivity, without overstimulating or inappropriately relaxing employees—thus possibly aligning the sensory qualities of workspaces with the functional demands of different settings

1.4 Research method

This study made use of various methods to answer the previously stated research questions. First an extensive literature study has been conducted to gain understanding of the current state of these methodologies and how they can be validated, plus the knowledge of psychophysiological stress and workplace design.

Then, the study made use of sensory ethnography as a methodology. The most appropriate specific execution of the method followed from the literature review. TU/e employees (PhD candidates) and students were asked to work at a different location for a certain period of time, during which the studied sensory modalities smell and touch of the environment were controlled, and the psycho-physiological stress was measured. More traditional psychology measurements (self-report) and physiological objective markers will be used as well, to compare with the ethnographic results and learn how the methods can possibly be combined or enriched one another.

1.5 Outline

The outline of this thesis is as follows. Chapter one introduces the topic and research objective. Chapter 2 provides a literature review on the topic. Chapter 3 explains the methodology and data collection. Chapter 4 provides a description and analysis of the data. Chapter 5 shows a discussion of the results and relates it to the existing literature. Chapter 6 concludes the report and provides future recommendations.

2 LITERATURE REVIEW

2.1 The known influence of workplace environment design on stress

The workplace environment is defined as the physical and functional setting encompassing all objects and stimuli employees encounter during work (Al Horr et al., 2016). Roskams and Haynes describe it as a complex psychophysical system, incorporating both the objective physical stimuli and the subjective perceptions of individuals within the space (Roskams & Haynes, 2020). The impact of the workplace on employees' mental health is influenced by their psychological and physiological responses to these environmental characteristics (Bergefurt et al., 2021; Schuller, in press). Workplace research on stress effects often focuses on explicit aspects such as workstation design, office layout, ergonomics, and interior decor, emphasizing elements like physical functionality and aesthetic design (Schuller, in press).

Physiological and psychological stress are interconnected but distinct concepts (Lupien et al., 2009). Psychological stress refers to emotional and cognitive reactions to situations when an individual perceives a situation as threatening and believes their resources are insufficient to cope (Becker et al., 2023). In workplace research this is often measured in self-reported/perceived stress and validated stress scale questionnaires capturing negative emotions such as stress, anxiety, tension or frustration (Becker et al., 2023). Examples of such questionnaires that are used in workplace design are (Short form) Perceived Stress Scale, State-Trait Anxiety Inventory, Job Stress Survey etc. (Bergefurt et al., 2021). Physiological stress, on the other hand, involves the body's physical reaction to stressors such as changes in the autonomic nervous system (Schuller, in press). This can be measured in many ways such as autonomic biomarkers (e.g. heart rate variability, electrodermal activity, blood pressure, respiratory rate) Endocrine biomarkers (e.g. cortisol, epinephrine, norepinephrine), immune biomarkers (e.g. cytokines) and neurological biomarkers (e.g. brain activity measured through EEG or fNIRS) (Schuller, in press). From these measurements, autonomic biomarkers, especially heart/pulse/respiratory rate, HRV (heart rate variability, and EDA (electrodermal activity) are used the most in workplace design research (Schuller, in press). Psychological and physiological stress differ in their concepts and how they are experienced and measured but are interconnected, as the prefrontal cortex links cognitive stress evaluation to physiological responses (Becker et al., 2023). These systems are related but do not always align, as evidenced by inconsistent correlations in empirical studies (Becker et al., 2023).

Bergefurt et al. (2021), in their scoping review, identified a significant body of research linking the physical office environment to employee stress. The most frequently studied building characteristics in this context include (day)light, noise/acoustics, office layout, and biophilic design, with stress assessed through both psychological and physiological measures (Bergefurt et al., 2021). Similarly, Al Horr et al. (2016) highlighted the influence of acoustic and biophilic factors in the workplace environment on employee stress. Schuller (in press) conducted a scoping review focusing specifically on the

physiological stress effects of the workplace environment and raised several concerns regarding the current body of research. While acknowledging the substantial volume of existing studies, Schuller (in press) noted a lack of consistency in outcomes, particularly when different physiological markers are compared. Additionally, the review pointed out a narrow focus on specific workplace characteristics, particularly biophilic design, with limited attention to tacit experiences and sensory mediation. This gap is significant, as sensory input plays a crucial role in the stress response (Lin et al., 2017) and can be directly linked to design considerations (Magsamen, 2019). For instance, the field of neuro-aesthetics describes the aesthetic experience as a complex interplay between biological sensory input and emotional reactions (Chatterjee & Vartanian, 2014). This perspective has gained traction within the built environment domain (Chatterjee & Vartanian, 2014; Magsamen, 2019). Consequently, there has been a growing call for more sensory-specific insights within environmental psychology, particularly regarding underexplored senses (Spence, 2020a; Schuller, in press), as well as the adoption of a multi-sensory approach (Bruno, 2022; Spence, 2020a).

The following section will explore the current knowledge surrounding the five primary sensory modalities, their effects on stress, and their role in shaping the physical workplace environment.

2.2 Sensory modalities and their potential relationship with psycho-physiological stress resulting from physical workplace environments.

Although there are more senses identified (Spence, 2020a), the five primary senses are considered: olfactory (smell), gustatory (taste), somatosensory (touch), auditory (sound) and visual (sight) (Comer et al., 2013). Each of these senses can play a crucial role in how we perceive and interact with our environment (Mallgrave, 2018), contributing to our psycho-physiological stress levels (Lin et al., 2017; Spence, 2020a). The sensory experience is produced by the integration of two different processes: sensation, the act of using the sensory system to detect environmental stimuli, and perception, the recognition and identification of those sensory stimuli (Comer et al., 2013). This can also be described with the concepts of bottom-up and top-down processing. Bottom-up processing starts with physical stimuli that proceed to transduce into neural impulses which are passed along towards more complex regions of the brain. Top-down processing is led by cognitive processes such as memories and expectations. These two processes are used simultaneously to perceive our environment (Comer et al., 2013). The senses influence the stress response by allowing the nervous system to detect and interpret environmental and physiological cues, coordinating systemic adaptations for survival (Lin et al., 2017). Sensory neurons act as sentinels, detecting threats like heat, toxins, or pathogens, and initiating stress responses by releasing neurotransmitters (e.g., serotonin), which regulate the stress response (Lin et al., 2017).

Yet, there is still not a complete understanding of how the five main senses play a role on the psycho-physiological stress of employees through the physical workplace environment. For instance, while visual and auditory inputs are well-documented (Al Horr et al., 2016; Bergefurt et al., 2022; Forooraghi et al., 2020; Kropman et al., 2023), the impacts of tactile, olfactory, and gustatory inputs on stress in the workplace have not been extensively studied (Spence, 2020a). This chapter will delve into the found influences the main sensory modalities have on the psycho-physiological stress levels of employees in workplace environments. Due to the more extensive work in vision and auditory experiences, and the aim of the thesis is to focus on lesser-researched sensory modalities, those chapters will provide a shorter summary of the existing body of work.

Table 1: A summary of the findings from the following literature section on the five main senses, their link to psycho-physiological stress and their link to the physical workplace design.

Sensory modality	Link to stress	Link to physical workplace design	Body of research
Vision	Vision influences stress by reducing physiological (Thayer et al., 2010) and perceived stress (Bergefurt et al., 2022) through exposure to natural environments, daylight, and well-chosen colour schemes, which help regulate cortisol levels and enhance mood. Whereas for example, clutter can increase stress (Forooraghi et al., 2020).	Vision plays a critical role by highlighting how elements like daylight, greenery, and colour schemes affect employee well-being, stress levels, and overall job satisfaction (Al Horr et al., 2016; Bergefurt et al., 2021; Forooraghi et al., 2020).	Large
Auditory	Auditory stimuli influence stress by causing physiological and emotional responses to unwanted noise, which can increase stress hormone levels, blood pressure, and overall stress (Ríos-Rodríguez et al., 2023). Music, however, could have some stress reducing effects (Chen, 2018)	Auditory stimuli play a role by highlighting the impact of noise levels and acoustic privacy on employee well-being, productivity, and stress, emphasizing the need for balanced soundscapes to optimize work environments (Al Horr et al., 2016; Contin De Oliveira et al., 2023; Kropman et al., 2023).	Large
Olfaction	Certain smells influence psycho-physiological stress, but effects are based on personal associations and opinions (Spence, 2020b). This could explain inconsistent findings (Liu et al., 2023).	Main focus has been on reducing mal odours instead of including positive smells (Spence, 2020b). Effects of positive smells are mainly tested in lab settings, not as much in field (Spence, 2020b)	Moderate, not as specific for office workplace design and stress

Somatosensation (touch)	Not widely researched, but there are indications it could have a possible effect, as there do seem to be links to emotions (Crippa et al., 2012) and preferences (Plaisier et al., 2023).	Not widely researched, the most somatosensation researched in the workplace is temperature (Pallasmaa, 1994; Paterson, 2016). Materiality and look or feel is researched, but often still from a visual, auditive or functional perspective instead of haptic (Al Horr et al., 2016; Colenberg & Jylhä, 2022; Fayyad et al., 2024).	Very little
Taste	There seem to be some connections between stress and taste, but mostly on stress influencing taste perception, not the influence of taste on stress (Ileri-Gurel et al., 2013; Nakagawa & Inui, 1997).	Taste is mostly related to facility management, not physical workplace design.	Very little but also low relevance for physical workplace design

2.2.1 Vision

Vision is one of the most researched sensory modalities in environmental psychology (Mallgrave, 2018; Spence, 2020a). Vision starts when light enters the eyes, falling onto the retina. Rods and cones photoreceptors change this light into neural impulses, which are then processed in the brain through the ‘what’ and ‘where’ pathway (Comer et al., 2013). The what pathway helps us identify an object, the ‘where’ pathway identifies its location. The interaction between the two helps us integrate complex visual stimuli. In the processing of the visual stimulus, top-down processing is involved a lot, to help us recognise and place meaning to what we see (Comer et al., 2013). As the effects of visual input on stress in the workplace environment are relatively well researched, this review will not dive into the details and provide an overview using mostly scoping reviews.

Within the workplace environment research, vision has had a rather centralised stance (Pink, 2013). Although, the visual perception is not always explicitly mentioned, the concepts that are researched lay their basis in visual cues (Bergefurt et al., 2021; Forooraghi et al., 2020). One example of these visual concepts are studies with real or virtual exposures to nature and biophilic design (Ríos-Rodríguez et al., 2023), where they either use biophilic design elements or create restorative break rooms to reduce the employee stress. Another example is the lighting and daylight (Bergefurt et al., 2021), where adequate lighting is researched for comfort and functionality, but also how it links to physical and mental health (Aries et al., 2015). In addition, more visually explicit research has been described for office environments, creating visually interesting office environments due to interior design features (Forooraghi et al., 2020), e.g. views, the use of wall decoration, variation in textures and colours, natural

materials, plants, look and feel, and the reduction of visual clutter (Al Horr et al., 2016; Bergefurt et al., 2021; Forooraghi et al., 2020).

Bergefurt (2022) found that when those papers researched mental wellbeing and psychological stress, this was often measured subjectively by asking the participants about their experience and feelings or using validated and not validated surveys. Several studies and reviews found reductions in perceived and self-reported stress due to biophilic design, nature elements and green spaces (Bergefurt et al., 2021; Berto, 2014; Shuda et al., 2020). Also, exposure to both daylight and artificial light plays a crucial role in reducing psychological stress and enhancing well-being (Bergefurt et al., 2021; Forooraghi et al., 2020). Increased daylight exposure has been linked to improved psychological health (Aries et al., 2015; Bergefurt et al., 2021). Additionally, higher illuminance levels, increased circadian stimulus (CS), and cooler light (higher correlated colour temperature, CCT) have been found to boost well-being, and psychological stress (Mills et al., 2007; van Duijnhoven et al., 2018). Furthermore, the look and feel of an office can influence psychological stress through aesthetics, colours, textures (often visually appraised), and spatial shapes (Kropman et al., 2023). Colours can affect mood (e.g., green promotes calm), textures trigger memory, and spatial design impacts comfort—open spaces aid creativity, while enclosed areas support focus (Al Horr et al., 2016). Aligning design with organizational values fosters belonging and reduces stress as well (Al Horr et al., 2016).

Physiological stress has also been related to visual concepts in office design (Schuller, in press). Several literature reviews and studies found exposure to nature elements and biophilic design reduces physiological stress, (Berto, 2014; Jones et al., 2021; Shuda et al., 2020). This has shown in several physiological measures such as cortisol levels (Jones et al., 2021), HRV, skin conductance, blood pressure (Ríos-Rodríguez et al., 2023; Shuda et al., 2020). Lighting and daylight too have been correlated with similar physiological stress markers (Al Horr et al., 2016; Aries et al., 2015; Thayer et al., 2010), where more exposure to daylight seems to have a positive effect on reducing physiological stress, however other artificial lighting subjects such as full spectrum effects still have varying results (Aries et al., 2015).

This is still a very short summary of the very extensive research within the visual aspects of workplace design and their effects on stress. Although there can still be some gaps in the literature, there is already a large basis to build on. This is why this thesis will not further elaborate on the effects of vision.

2.2.2 Auditory

Hearing is the process of detecting, transmitting, and interpreting sound waves through the auditory system (Comer et al., 2013). Sound waves cause eardrum vibrations, which are amplified by the ossicles and transmitted to the cochlea, where mechanical energy is converted into electrical signals for processing in the brain (Pickles, 2012; Comer et al., 2013). Perception involves interpreting pitch, loudness, timbre, and localizing sounds, influenced by cognitive and emotional factors (Moore, 2012).

Primarily, hearing serves to alert and warn, triggering reflexive responses and stress reactions. Over time, it evolved to support communication and entertainment as sound gained symbolic meaning (Westman & Walters, 1981). In environmental psychology research, it is important to make note of the distinction of sound and noise, where noise is inherently negative and unwanted sounds, but sound can encompass both negative and positive auditory experiences (Comer et al., 2013).

Within workplace environment research, the role of sound has been widely explored, particularly in relation to noise exposure. In office environments, noise is consistently found to be a stressor (Al Horr et al., 2016; Bergefurt et al., 2022; Colenberg et al., 2021; Contin De Oliveira et al., 2023). More specifically, scoping reviews on open office plans indicate that human-induced sound, such as conversations, is particularly stressful and annoying (Contin De Oliveira et al., 2023). One explanation for this is privacy concerns, as background noise, especially from human voices, affects perceived privacy, productivity, and well-being (Contin De Oliveira et al., 2023). Another explanation can be that the brain picks up particularly intelligible conversations and is thus distracted by them (Haapakangas et al., 2014). Studies suggest that higher rates of acoustic privacy correlate with increased productivity and well-being (Kropman et al., 2023). However, when examining healthy and preferable office conditions, less noise is not necessarily more desirable or less stress-inducing. Two scoping reviews on indoor office quality and psychological effects suggest that a vibrant work environment with moderate background noise is generally preferred, even though excessive or disruptive noise can be detrimental to mental well-being, leading to increased stress and decreased productivity (Al Horr et al., 2016; Bergefurt et al., 2021). This indicates a need for balance: while some background noise can contribute to a dynamic work atmosphere, excessive noise levels can cause stress. Although it is researched a little less frequent, there is also research regarding more positive sounds in the workplace environment. For example, including ‘booster breaks’ during office hours during which employees listened to nature sounds was suggested to have positive effects on their stress levels (Largo-Wight et al., 2016). Also the effects of listening to music in a work environment has been studied and show some indications it might have stress reducing effects (Chen, G., 2018; Nomura et al., 2005), yet there are still difficulties finding consistent results (Nomura et al., 2005), and the research domain is often more focussed on healthcare workplaces (Raglio et al., 2020).

As hearing primarily has served people to alert and warn, it is inherently related to our emotions and psycho-physiological stress response, depending how much the sound is perceived as a threat/ stressor (Westman & Walters, 1981). The focus of stress in the workplace and noise, too seems to lie mainly with psychological stress via self-report (Bergefurt et al., 2021). A scoping review on the effects of noise on health found that noise annoyance is a widespread issue causing stress-related symptoms such as anger, displeasure, exhaustion, and general stress (Basner et al., 2014). Also, the effects of nature sounds were measured along the measure of self-reported psychological stress, indicating a decrease in psychological stress when employees were exposed to short breaks of nature sounds (Largo-Wight et

al., 2016). Although it is important to note that high variation in nature sounds can again increase disruption and psychological stress (Newbold et al., 2017). Music too was measured along psychological stress (Chen, G., 2018), showing varying results with for example differentiating due to type of tasks, type of music or personal preferences (Nomura et al., 2005; Raglio et al., 2020).

Physiological stress has also been linked to auditory stimuli in workplace design. Noise exposure has been associated with increased physiological stress responses, such as elevated cortisol levels, heart rate variability (HRV), skin conductance, and blood pressure (Basner et al., 2014; Ríos-Rodríguez et al., 2023). Prolonged exposure to high noise levels can contribute to chronic stress, cardiovascular issues, and sleep disturbances (Al Horr et al., 2016; Aries et al., 2015; Thayer et al., 2010). On the other hand, certain auditory stimuli, such as nature sounds or controlled background noise, have been linked to reduced physiological stress responses, suggesting potential applications in office environments to mitigate negative effects (Berto, 2014; Shuda et al., 2020).

This review highlights the extensive research on auditory aspects of workplace environments and their effects on stress. While noise is a well-documented stressor, there are still some gaps in the effects of positive and beneficial sounds on the workplace. Yet due to the extensive body of research that already exist, this thesis will not further discuss auditory effects.

2.2.3 Olfaction

The process of olfactory sensation begins when odorants, airborne chemicals, enter the nose and interact with sensory receptors in the nasal mucous membranes (Comer et al., 2013). Odorants bind to these receptors in a lock-and-key mechanism, where specific chemicals fit particular receptors. When sufficient odorant molecules bind to the receptors, they trigger an action potential in the olfactory receptor neuron, sending a neural signal to the brain. Continuous exposure to the same odorants can lead to nose blindness, where the perception of the odour fades over time of exposure (Comer et al., 2013). This exploration of olfactory function includes the environmental stimuli that produce smells, the sensory organs involved, and the conversion of these stimuli into brain signals. Additionally, it examines the development of olfactory abilities, individual differences in these senses, and potential dysfunctions in these sensory systems (Comer et al., 2013). Where other senses first travel to the thalamus, the ‘switchboard’ in the brain relaying information to the other parts of our brain, smells directly travel to our limbic system, including the amygdala and hippocampus (Bentley et al., 2022). This is where we process emotions, learn and create memories. This creates a strong connection between smells and our emotions and memories (Bentley et al., 2022).

The olfactory atmosphere of a built environment plays a significant role in shaping experiences, emotions, and behaviours (Bentley et al., 2022; Liu et al., 2023). While historically, urban environments were filled with unpleasant odours, modern design aims to eliminate malodours and sometimes introduces pleasant ambient scents for branding or well-being purposes (Spence, 2020b). However, the

role of smell is often overlooked in architectural theory, despite its ability to enhance the multisensory experience of a space (Spence, 2020b). Offices have largely focused on eliminating unpleasant odours rather than incorporating pleasant scents for well-being. Poor air quality and the presence of unfamiliar odours in enclosed spaces have been linked to sick-building syndrome (Spence, 2020b). But there is some research suggests that scenting office environments could improve mood, reduce stress, and enhance cognitive performance, though findings remain inconsistent (Huang & Capdevila, 2017). Specific scents such as bergamot and petitgrain essential oil have shown potential in reducing workplace stress and improving task performance (Huang & Capdevila, 2017; Liu et al., 2023). Spence (2022b) notes that a dynamic approach to scenting offices—where scents are changed periodically—might prevent associations between specific scents and workplace stress, which could otherwise develop over time.

As smell is very closely linked to our emotions and memories, it can have quite strong effects in psychological stress, as well inducing as reducing (Bentley et al., 2022; Spence, 2020b). The associative nature of smell means that certain odours, like the scent of a dentist's office (e.g., clove oil), can induce stress and anxiety (Spence, 2020b). Pleasant scents, such as nature scents (conifer tree), citrus or lavender, have been tested as potential stress reducers (Bentley et al., 2022; Matsubara et al., 2023). Also scents of orange and bergamot have been found to have a psychological stress-reducing effect in stressful situations (Liu et al., 2023). However, their effectiveness on psychological stress depends on personal associations and context (Matsubara et al., 2023; Spence, 2020b). Spence (2020b) emphasizes that scent is only one part of a multisensory experience, and incongruency between sensory inputs (e.g., a calming scent in a visually chaotic space) may reduce the overall effectiveness of olfactory interventions or can even be uncomfortable and stress-inducing (Spence, 2020b; Fenko & Looock, 2014). Fenko and Looock (2014) also mentioned that the combination of multiple sensory interventions could lead to lower stress reduction or even be stress inducing due to overstimulation.

There is some research suggesting that smells can be used to make a positive influence on the psychophysiological stress of employees in the workplace. Yet, results are not yet very consistent, possibly due to interpersonal associations (Spence, 2020b). There seems to be variation over psychological and psychological stress (Liu et al., 2023). Some other studies too argue physiological stress effects. Certain scents can trigger autonomic responses, such as changes in electrodermal activity and heart rate variability (Huang & Capdevila, 2017). Studies indicate that essential oils, particularly from Cupressaceae trees, can increase parasympathetic nervous system activity, leading to relaxation and stress reduction (Matsubara et al., 2023). Similarly, petitgrain essential oil has been found to balance autonomic nervous system activity, improving work performance while reducing workplace stress (Huang & Capdevila, 2017). But there is still a lot of information lacking in the field such as more in-field studies, long term studies, multi-sensory effects, mechanisms and delivery studies (Spence, 2020b). Other research has demonstrated that there is a potential disconnect in the effects of scents on

psychological stress compared to physiological stress (Liu et al., 2023). Liu et al. (2023) found that the self-reported stress was lower in offices with bergamot scent (for both men and women), yet the physiological stress, based on HRV, was increased for men. They report that differentiation in psychophysiological stress is not uncommon (Liu et al., 2023). Amano et al. (2012) did find similar results between psychological (self-reported) and physiological (ECG readings) effects, but they measured the effects of scents that were preferred by participants. This preference indication also reads into the idea of Spence (2020b) that the effects of stress might really depend on your personal associations.

2.2.4 Somatosensory (Touch)

Touch refers specifically to the sense of physical contact with the skin, detecting pressure, temperature, and texture. Tactile sensation is a specific aspect of touch that deals with detailed perception of textures and fine physical details through the skin. This chapter will mainly focus on the aspects of touch. The touch system is an intricate sensory network responsible for the perception of touch. It involves various sensory receptors within the skin that detect different types of physical stimuli such as pressure, pain, temperature, details and textures and vibrations (Comer et al., 2013). The sensitivity of different body parts to touch varies, largely due to the density of free nerve endings in those areas. More sensitive regions, have a higher density of these nerve endings compared to less sensitive area (Comer et al., 2013). When touch occurs, free nerve endings send signals through the spinal cord to the brain. These signals are first received by the thalamus and then transmitted to the somatosensory cortex in the parietal lobe for processing (Comer et al., 2013). Sensory adaptation can reduce the sensation of a constant tactile stimulus over time, such as the feeling of clothes against the skin shortly after getting dressed. This phenomenon highlights how our sensory system adjusts to continuous stimuli to prevent sensory overload (Comer et al., 2013). Overall, the tactile system's complex array of receptors and neural pathways enables the detailed and nuanced perception of touch, pressure, temperature, and pain, playing a crucial role in how we interact with our environment.

Touch plays a crucial role in shaping the experience of the built environment, including workplaces such as offices. While architecture often prioritizes visual aesthetics, the tactile qualities of materials and surfaces influence how individuals interact with and perceive their surroundings (Kraftl & Adey, 2008; Paterson, 2016). Buildings and spaces are not only seen but also felt and experienced through physical interaction (Paterson, 2016). Kraftl and Adey (2008) argue that architecture is not just about visual aesthetics but also about the tactile and haptic experiences it provides. They mention the importance of considering the affective, tactile, sensual effects of buildings, which often evade purely symbolic or visual interpretations. By focusing on the material, performative, and embodied aspects of architecture, designers can create environments that enhance the inhabitant's experience, making the built environment more interactive and meaningful (Kraftl & Adey, 2008). There are studies that focus on the materiality of indoor work environments, and even how they relate to a healthy work environment

(Colenberg & Jylhä, 2022; Fayyad et al., 2024), but these too seem to judge the materials through a visual (aesthetic), auditive (acoustic) or functional (easy to clean) lens. The same is held for the terms ‘look and feel’, there are some studies that investigate those aspects of a workplace environment (Al Horr et al., 2016; Bergefurt et al., 2022; Kropman et al., 2023). Although the word feel is generally related to the word touch, in this research context it is often described from a more aesthetic and social point of view *‘The definition of the ‘look and feel’ of a workplace is subjective and depends both on social norms and personal preferences. A good office design would aim to understand and address an organisation’s aesthetical outlook and values in the workplace design strategy’* (Al Horr et al., 2016). The pure tactile aspects are still often rather neglected in architecture (Pallasmaa, 1999). The main haptic functions we see in our modern day built environment is mainly to help the visually impaired with street safety and wayfinding. Herssens and Heylighen (2012) identified three surface categories: Movement planes (floors and paths for guiding movement), Guiding planes (walls and handrails for orientation), and Rest planes (seating and leaning surfaces for support). They also distinguished between active, dynamic, and passive touch and defined key haptic qualities like texture, hardness, and temperature. Smooth surfaces were preferred for movement, rough textures aided guidance, and material properties influenced comfort. Taking these factors into account during designing can enhance spatial experiences for all users, especially for those relying on touch for navigation. These types of touch and surface categories have not yet been translated to the workplace environment. Although the furnishing of an office is often well thought of, it is often visually based, even though the touch of elements such as furniture, flooring, and work surfaces can subtly affect comfort, engagement and overall well-being (Herssens & Heylighen, 2012).

As touch is not very much regarded in environmental psychology, there also is no research concerning the effects on psychological stress in that context. Yet there are some indications that touch could influence emotional responses and affective states in other fields of research. Research from product design suggests that materials evoke emotions, with natural-feeling materials often associated with positive emotional responses (Crippa et al., 2012). Zheng and Morell (2012) found that haptics significantly affect the affective state, indicating that the actuation profile (tactile sensation patterns made by haptic devices), body site, actuation type, intensity, and geometry all play a role and found that symmetric pressure actuation profiles are more relaxing than asymmetric ones (Baumann et al., 2010). Furthermore, smooth and natural textures are generally perceived as pleasant, while irregular or artificial surfaces may contribute to discomfort (Plaisier et al., 2023). Also, Herssens & Heylighen (2012) showed preferences and higher comfort among the blind for certain haptic design choices. In the literature concerning the other sensory modalities, there often seemed to be a link between personal preferences/pleasantness and mood and self-reported stress, such as preferred and positive smells (Spence, 2020b), music (Raglio et al., 2020) and other sounds or views that are considered nice (Bergefurt et al., 2021). These preferences and other links to moods could suggest that there might be a

similar effects in touch preferences/pleasantness in the physical environmental. It could suggest that for example textures that are rated as more pleasant, could reduce stress or improve the mood in a similar way.

Considering physiological stress there is even less research. There is one study on haptic stimulation suggesting that certain tactile sensations, such as slow and rhythmic patterns, can induce calming effects by influencing heart rate and physiological arousal (Xu et al., 2021). They have performed a study exploring how haptic stimulation, mimicking slowed-down heartbeats, influences emotional responses during stressful tasks, particularly considering individuals' interoceptive accuracy (IAc), which is their ability to sense internal bodily signals. They found that haptic stimulation simulating slowed-down heartbeats modulates one's physiological state, depending on their interoceptive accuracy (Xu et al., 2021). Yet, this is very far from environmental psychology. But just because this has been unexplored does not mean that there are no effects. The indication that there might be psychological stress effects, suggest that there could also be physiological stress effects, even though they are very different, they are also still linked (Lupien et al., 2009). While these effects have been observed in controlled experimental settings, their application in office environments remains largely unexplored.

The effects of touch in environmental psychology, especially in the workplace concerning psychophysiological stress is very much unexplored. But there are some indications that future research and design practices could benefit from considering the haptic qualities of materials to create workplaces that not only function efficiently but also support employee well-being through subtle but impactful tactile experiences.

2.2.5. Gustation

Taste, or gustation, is a complex sensory system that allows humans to perceive and differentiate flavours. It involves the detection of chemical substances by taste receptor cells located on taste buds, primarily on the tongue but also on the soft palate and other areas of the mouth and throat (Comer et al., 2013). Each taste bud contains 50-100 taste receptor cells, which are specialized to detect one of the five primary taste modalities: sweet, sour, salty, bitter, and umami (savory) (Briand & Salles, 2016). When these substances interact with the taste receptors, they trigger a series of biochemical events that generate electrical signals transmitted to the brain via the gustatory nerves (Briand & Salles, 2016). The brain processes these signals in the gustatory cortex, integrating them with other sensory inputs such as smell and texture to form a complete sensory experience (Comer et al., 2013).

Taste seems to have a two-way dynamic relationship with stress. Stress has a proven effect on the taste perception, decreasing the sweet taste sensitivity and increasing the bitter taste sensitivity (Ileri-Gurel et al., 2013; Nakagawa & Inui, 1997). Research on the effects of taste on stress are still rather lacking. Although this could be an interesting research area, taste and food in the workplace mostly concerns

facility management provision. It is therefore not really relevant for researching the physical workplace design. This is why taste will be excluded from further investigation in this thesis.

2.2.6. Concluding all senses:

In conclusion, while all five senses—vision, sound, touch, taste, and smell—are shown to exhibit some influence on stress, their impact is predominantly linked to nature-based stimuli, aligning with the well-documented nature effects by Ulrich (1991) and Kaplan (1995). Despite the broad scope of sensory experiences, workplace environmental research has traditionally focused on vision and sound as the primary modalities influencing psycho-physiological stress. The body of research concerning those modalities is large, with extensive and appropriate research methods already explored and verified. The taste modality, is less researched but is more closely related to workplace food and beverage options, which stem from facility management rather than the physical environment itself. Given these distinctions, this thesis will exclude the senses of vision, sound, and taste, and instead concentrate on touch and haptics. These modalities show promise, but are not researched as extensively. Future research in this field could be interesting, but appropriate and in-depth methodologies are still necessary to be able to explore those. This is why this thesis narrows down the focus to touch and smell and aims to provide deeper insights into what can we learn from sensory ethnographic methods to include more touch and smell to improve research on the workplace design-physiological stress relationship.

2.3 What are sensory ethnographic methods

2.3.1 What is ethnography and how is it used in workplace research and in practise?

Ethnography involves integrating various information sources to gain deeper insights into a particular group or culture, capturing actions, facts, and behaviours to create a representative overview (Hampton et al., 2019). According to John Brewer (2000) ethnography is the study in naturally occurring settings or fields through methods that capture social meanings and everyday activities, with researchers participating directly in the setting to systematically collect data without imposing external meanings. Gobo (2011) describes ethnography as based on direct observation, supplemented by listening to conversations, reading documents, and asking questions, with a distinctive emphasis on active cognitive modes of observing, watching, and scrutinizing. Pink (2013) mentions how the involvement of the ethnographer adds a new layer of reflexivity, as well in the questioning as in the results, which can lead to deeper and more pinpointed questions and more insightful findings. In the workplace environment this can be very useful as many of the underlying mechanisms and thoughts are not always in the front of our minds, and therefore can be missed easily.

2.3.2 What is sensory ethnography and what can it teach us?

Sensory ethnography is an interdisciplinary approach that integrates ethnographic research with sensory-based techniques to explore human experiences of the senses, using typical ethnographic methods such as interviews as well as more sensory specific methods such as techniques from perfume makers (Pink, 2015). It emphasizes sensory perception—sight, sound, touch, taste, and smell—as crucial to understanding the world. Researchers engage in immersive fieldwork to examine how sensory stimuli shape behaviours, emotions, and social interactions, providing a nuanced and multi-dimensional view of culture and society as well as how those aspects influence the sensory perception (Pink, 2015; Zickar & Carter, 2010). These are insights that are difficult to measure using for example biomedical technology, which makes sensory ethnography an interesting avenue to consider the role of sensory perception in mediating the physical workplace environments and experiences such as psycho-physiological stress (Pink, 2015). Methods range from common techniques like observations and sensory diaries (Pink, 2015) to less conventional ones such as smell-walks (Adams et al., 2008; Low, 2015), participant-led photography (Pilcher et al., 2016), and sensory mapping (McLean, 2020b).

Ethnographic methods are widely applied in workplace research (Malachowski et al., 2016; Tagliaro et al., 2023; Zickar & Carter, 2010). Initially valued for capturing workplace realities, ethnography's role diminished as researchers prioritized quantitative data (Zickar & Carter, 2010). However, its importance persists, offering deeper insights into participants' perspectives and acknowledging researchers' assumptions (Zickar & Carter, 2010). Workplace ethnography takes various forms, including affective ethnography, which examines emotions and bodily experiences (Holck & Villeseche, 2023), autoethnography, which reflects the observer's standpoint (Orel, 2023), and digital ethnography, which explores experiences in digital work environments (Ritter, 2023). Institutional ethnography further investigates how everyday workplace activities connect with broader institutional structures (Malachowski et al., 2016; McGibbon et al., 2010). While workplace ethnography is established, sensory ethnography remains underutilized in the workplace, primarily appearing in consumer research, healthcare, user-centred design, and the arts (Pink, 2015). However, its potential relevance increases as non-visual and non-acoustic sensory modalities gain recognition.

Sensory ethnography offers a valuable approach to studying workplace stress by immersing researchers in employees' multisensory experiences. Traditional ethnographic studies have relied on observation and interviews to explore workplace stress (Malachowski et al., 2016; McGibbon et al., 2010). Sensory ethnography extends these methods by capturing embodied aspects of stress—such as how specific environments, sounds, or textures contribute to or alleviate stress (Pink, 2013, 2015). Techniques like audiovisual recordings can document sensory stimuli linked to stress responses, while participatory methods such as sensory mapping allow individuals to track sensory stressors throughout their day (Pink, 2015). Additionally, observational approaches using behavioural indicators, such as body

language (Dael et al., 2012a, 2012b) or stress-related facial expressions and gestures (Koldijk et al., 2018), provide objective insights beyond self-reported data. These methods enhance our understanding of stress by capturing unconscious or unarticulated experiences.

In the following chapters, this thesis will elaborate on several sensory ethnographic methodologies, and how they could be applied to research the experiences of smell and touch, and how they can be applied in the workplace environment to research psycho-physiological stress. Table 2 below provides a short overview of the discussed methods.

Table 2: A summary of the findings from the following literature review section on several most-used sensory ethnographic techniques, and how they can be used to evaluate the touch and smell experience.

Method	Explanation	Evaluating touch and smell
Observations	Researchers immerse themselves in a community, engaging in multisensory participation to understand the sensory experience through everyday activities like eating and walking (Howes & Classen, 1991; Pink, 2015).	Engaging with participants in sensory walkabouts or everyday activities, where they narrate and discuss their sensory experiences, allowing the ethnographer to explore these senses in real-life contexts (Low, 2015). Mostly done for smell, but touch not so much yet.
Interviews	Researchers incorporate participants' sensory experiences by conducting interviews in meaningful environments and using sensory prompts, such as sounds, images, or objects, to evoke deeper insights into their lived experiences (Harper, 2002; Pink, 2013).	For touch, the prompts can be used as well as in situ interviews where participants are asked to interact with the environment (Paterson, 2009). For smell in situ interviews can be used (Harris & Guillemin, 2011).
Sensory Diaries	This is a self-report ethnographic method where participants regularly document their sensory experiences in real-time, offering insights into the nuances of daily life and personal sensory perceptions in natural contexts (Bolger et al., 2003; Reid et al., 2011). Either in 'free of direct instruction' style, or via a questionnaire (Reid et al., 2011).	Researchers can direct participants to regularly document their smell or touch experiences in situ or from memory, such as the intensity of aromas or tactile sensations, along with emotional responses in real-life contexts et al., 2012; Rodrigues et al., 2021).
Participant-led-photography	Participants are asked to take photographs to document their experiences and environments, often followed by interviews to	Participants can be directed in taking photographs from touch experiences, smells are more difficult to capture as they are not

	explore the meanings and emotions behind the images, providing deeper insights into their lives (Burles & Thomas, 2014; Warren, 2002).	visible (Pink, 2011). Yet, a photograph from the expected source could already invoke memories in the interview (Latham, 2003).
Sensory mapping	Participants are asked to document their sensory experiences, such as smells, sounds, and tactile sensations, on a map, visually representing how sensory stimuli shape perceptions of and attachments to specific places (McLean, 2020b; Misharina et al., 2023).	Participants record the locations and characteristics of tactile or olfactory experiences in an environment, with smell being commonly explored through smell-walks (McLean, 2020b) and touch requiring more guidance to capture intentional tactile interactions (Misharina et al., 2023).

2.3.3 Observations (through participation)

In classic participant observation in sensory ethnography, as discussed by Howes and Classen (1991), the researcher goes along with the participants while asking questions. They for example work along the employees of a to be researched companies for a certain amount of time, mostly aiming to understand the cultural and social systems of the researched group and try to see the surroundings. However, they often overlook the experiential aspects of ethnography, focusing predominantly on visual observation (Pink, 2015). To negate this, Pink (2015) proposes a more phenomenological approach to participation in sensory ethnography, which emphasizes understanding experiences as multisensory, not just visually dominant. This method involves actively engaging through the ethnographer's own sensory experiences, beyond mere observation and interviews, to embody and share the sensory worlds of others (Pink, 2015). Ethnographers are encouraged to become emplaced within the communities they study, co-participating in the creation of place and understanding, thereby gaining deeper insights through multisensory immersion and interaction. She suggests that the ethnographer can learn through participation, particularly by focusing their engagement in the more commonplace and everyday activities such as eating and walking together with the participant, and sharing the experiences (Pink, 2015).

How does it relate to measuring the sensory modalities: smell and touch?

In the western ethnographic observations concerning sensory modalities, the focus has been very vision dominant (Howes et al., 1991; Howes & Classen, 1991; Pink, 2015; Sparkes, 2009). Yet, the other sensory modalities are not completely ignored, and observational methods have been used to study them to a certain extent. For smelling, often in an urban setting, this is often done in a walkabout (Adams et al., 2008; Low, 2015; Sparkes, 2009). These sensory walkabouts involve asking participants to narrate their sensory experiences while walking urban environments or specific areas (Low, 2015). This method

provides a vivid exploration of sensory encounters within everyday spaces, shedding light on the multifaceted nature of sensory experiences in different contexts (Pink, 2015). For example Low (2015) describes having performed smell walkabouts through the city of Singapore to investigate the smellscape (olfactory environment). He walked together with the participants, experiencing and discussing the smells in the context. He argued that this approach allowed us to transition from discussing imagined smells in narrative interviews to experiencing actual in situ smells that respondents identified within specific contexts (Low, 2015). For the auditory and visual senses this can be aided by recording technology, but for smell this is more difficult, as smell is not 'recordable'. Similar methodologies are used for the observational ethnographic research on smells, often in urban settings.

Although it is not researched as often, similar methodologies could be used examining touch. It might not need to be a walk per se, but as Pink (2015) noted, an everyday, common experience where the ethnographer experiences the haptic senses together with the participant in situ.

Is it used within the workplace to measure design aspects?

Sensory observation through participation is akin to the methodology of autoethnography (Pink, 2015). Autoethnography is a methodology that allows the ethnographer to use their own experiences to produce academic knowledge. This methodology is often used in workplace research (Orel, 2023). Although there are more layers and methods to autoethnography, observation and analysing their own experience is one of them (Orel, 2023). McGibbon et al. (2010), for example, used it in a hospital workplace setting, not as much on the sensory aspects, but more institutional. The method was combined with other methods such as alongside interviews and focus groups, to measure the stress amongst nurses (McGibbon et al., 2010). As was described above, the smellscape observations are often performed in the form of walks, often on an urban scale (Low, 2015). But this does not have to be limited to this urban scale. One could imagine a walk throughout the office building as well, or more in the traditional sense where participant and ethnographer work alongside each other for a certain amount of time, where smell of haptic experiences can be asked about at multiple instances when interacting (Howes & Classen, 1991; Pink, 2015). This is not yet a common method but does have the potential to be used more.

Is it used to measure psycho-physiological stress?

The method is inherently based on self-report and qualitative measures. Therefore, most of the observations are based on psychological stress. Observations can be made using the method of observation through participation, as Pink has described in her book (Pink, 2015). There are very few sensory ethnographic practices that specifically measure stress, but there are other ethnographic observations that do measure stress. This is for example done by McGibbon (2010). Although this is not sensory ethnography, they did aim to measure stress in the workplace amongst nurses, also using

interviews and text analysis. They engaged in the daily activities and observed their everyday and every night work. During these observations, as well as during the interviews, they listened to the stories of the nurses. Questions they tried to pay attention to and answer were for example: ‘*What words did the nurses use to describe their experiences? How did models of nurses’ stress tend to silence these standpoints?*’ (McGibbon et al., 2010). This is not sensory, but it might be possible to link it to sensory aspects as well. These kind of questions could be altered towards the sensory experience, for instance on a smell walk.

There are also non-verbal indicators of behaviour amongst participants that might show stress, that could be noted during observations. These can provide valuable insights when certain emotions are not recognised and/or verbalized. Mohiyeddini and Semple (2013) identified displacement behaviours linked to stress, such as grooming, hand-face contact, yawning, fumbling, and lip-biting. Fidgeting, including object manipulation (e.g., pens), is also a common stress response (Woodward & Kanjo, 2021). Their fidget cube, using deep learning classifiers, offers an objective tool to assess and mitigate stress. Dael et al. (2012a, 2012b) developed the Body Action and Posture Coding System (BAP) to link body movements to emotions, while Koldijk et al. (2018) created an automatic classifier to infer workplace stress from multimodal sensor data. Their study found posture most effective in distinguishing stress levels, with facial expressions enhancing accuracy. Though these observations are not strictly *sensory* ethnographic, they could complement such methods with additional and maybe more objective insights.

2.3.4 Interviews

Sensory ethnography, as discussed by Pink (2015), expands traditional interviews by integrating sensory experiences, providing deeper insights into participants' lived experiences (Calvey, 2021). Instead of relying solely on verbal communication, this approach encourages active engagement with sensory environments. One method is conducting interviews in meaningful or research-relevant settings, allowing both interviewer and participant to remain active in the environment. This helps capture how sensory contexts influence experiences. Rather than being seated and stationary, interviews can take place while moving through a space, fostering a more natural and immersive discussion (Pink, 2015). Another approach is mediated interviews, where sensory prompts—such as images, sounds, textures, or personally significant objects—help participants recall and articulate experiences in greater detail (Harper, 2002; Pink, 2013). This method transforms interviews into reflective and interactive processes, enhancing memory recall and encouraging richer descriptions of sensory perceptions. Interviews are often used alongside other ethnographic methods such as observations (McGibbon et al., 2010) and participant-led photography, which will be elaborated on in a later chapter (Harper, 2002; Latham, 2003; Pilcher et al., 2016). Conducting interviews in meaningful settings and using sensory elicitation can enhance understanding of participants' sensory subjectivity and contextual meanings

(Pink, 2015). Since knowledge, practices, and experiences are shaped by personal and cultural factors—and individuals shift between subjectivities—researchers must critically reflect on their role in knowledge production.

Harris and Guillemin (2014) demonstrated the benefits of sensory ethnography in healthcare research by asking participants to describe sensory elements like smells and sounds, uncovering aspects of experience often overlooked in traditional interviews. By capturing a full range of sensory information, these methods provide rich, immersive data that reveal how individuals perceive and interact with their surroundings. They also make interviews more engaging and meaningful for participants, as they reflect comprehensively on their lived experiences (Guillemin & Harris, 2014; Pink, 2015). However, sensory interviews can be complex and time-consuming, and the subjectivity of sensory experiences makes data analysis challenging, particularly when participants freely define sensory prompts (Calvey, 2021).

How does it relate to the sensory modalities: smell and touch?

Smell has been proven to have a strong effect in recollection of memories, which would make smell elicitation a useful method to help recall memories (Harris & Guillemin, 2011). Although interview questions such as ‘What did that place smell like?’ or ‘Is there a smell that particularly reminds you of that time?’ (Guillemin & Harris, 2014) can be asked, elicitation might be a little bit more difficult. Smells are difficult to capture, bring with, visualise or control, especially when researching the smells of a certain area (compared to a perfume) (Pink, 2015). There are examples of scratch-n-sniff studies where participants were asked to answer questions on experience and induced memories, which could be translated into an interview setting (Guillemin & Harris, 2014; McLean, 2020b). Yet, this is a rather complex method to execute. The most effective way might be to place the interviews in the studied environment, more in an observational/participatory way (Pink, 2015). This way, the smells are very present and can help elicit the memories. Unfortunately, this still comes with the sidenote that the smells are still difficult to control. An area might smell differently on different days or even different times of the day.

Touch too is a very important sense in the recollection of memories, which can be used in interviews as well as by using the materiality of an object, as through a line of questions and elicitation that tap into the tactile sensations such as movement and texture (Paterson, 2009). These elicitations can be in the form of objects as well as photographs of the objects and environments that is in the research objective. Even though, the use of photographs or other visual elicitation as drawings is inherently a visual task, the photographs are not just visual images, but contain other sensory information as well (Guillemin & Harris, 2014; Pink, 2015). For example the materiality of objects in the photograph, participants might remember or imagine what it might feel like, through embodied touch (Mallgrave, 2018). These prompts do not necessarily need to be preselected by the researcher but could also be indicated during the interview encounter, initiated by either the interviewer but also by the interviewee

(Harris & Guillemin, 2011). Deriving from this, it would be important that the location of the interview would be set in the to be studied context and environment.

Is it used in the workplace to measure design

Interviews are widely used in office design research but often follow a traditional, stationary, talk-centred format (Pink, 2015). Pink argues this limits understanding, as sensory experiences are also expressed through gestures, facial expressions, and touch. Interviews are also used in the new trend of participatory design, often combining interviews with workshops (Markkanen et al., 2022), but the interviews remain largely verbal. Similarly, workplace focus groups note body language and tone but often overlook elicitation methods or environmental context, potentially losing valuable insights (Golvani et al., 2021).

Pink (2015) emphasizes that interviews should take place in meaningful settings, such as participants' desks or while walking through the office, to reveal sensory insights like seating comfort or airflow. When on-site interviews are not possible, elicitation methods—such as object or photo prompts—can enhance engagement. Warren (2008) used photo-elicitation in design aesthetics research, incorporating participant-led photography to capture emotions and storytelling, recognizing that aesthetics are difficult to express through language. She also emphasized understanding participants' cultural backgrounds, aligning with Pink's (2015) view on the importance of first researching the (inter-)subjectivity of both the researcher and the participant. So although interviews are often used in workplace design, only few instances this is done in a rich ethnographic way as described by Pink and Warren.

Is it used to measure psycho-physiological stress?

Interviews, though not widely used as a primary method for studying stress in workplace design, have proven to be valuable in exploring stress-related experiences. According to Bergefurt et al.(2021), only a small percentage of research on stress in workplace design utilizes interviews. However, interviews, particularly focus groups, have yielded important insights. For instance, Golvani et al. (2021) examined the experiences of nurses working in no-daylight environments through focus group interviews. Although stress was not the primary focus of their study, open-ended questions like “What is your experience regarding limited access to daylight during a workday?” revealed that heightened stress was frequently mentioned by participants, which was uncovered in the thematic analysis, and described in those themes as result.

In contrast, a study by McGibbon et al. (2010) directly focused on stress, combined interviews with other ethnographic methods such as observations and focus groups. Their in-depth interview questions, such as “Tell me about some of the challenges in your work?” and “How do you manage these challenges?” provided insights into the personal and professional factors contributing to stress

(McGibbon et al., 2010). The researchers went beyond the literal answers, examining the language nurses used to describe their experiences and considering how dominant models of stress might suppress alternative viewpoints (McGibbon et al., 2010). This approach aligns with Pink's (2015) view that interviews should focus not only on spoken words but also on the nuances of expression and context. By incorporating participant observation, McGibbon et al. gained deeper insights into stress in the nursing profession (McGibbon et al., 2010).

2.3.5 Sensory diary

A diary is a sequential documentation of participants' activities and experiences, capturing ongoing social, psychological, and physiological processes in everyday contexts (Bolger et al., 2003). As self-report tools, diaries provide valuable insights into daily life by recording small yet significant experiences (Bolger et al., 2003). They are widely used across psychology, healthcare, product development, and food research (Chen et al., 2012; Jacelon & Imperio, 2005; Reid et al., 2011; Rodrigues et al., 2021). These diaries are often question-prompted, and Bolger and colleagues (2003) emphasize the importance of aligning the research question with the research design.

Diary studies typically follow three protocols: interval-contingent (regular reporting at fixed intervals), signal-contingent (entries prompted by random or fixed signals), and event-contingent (entries triggered by specific events) (Bolger et al., 2003). Diaries can also be easily used alongside other methods such as diary-interviews and diary-photography (Latham, 2003). In diary-interviews, participants first document their daily experiences, thoughts, and feelings, which are later explored in-depth during an interview. Adding disposable cameras allows participants to visually capture meaningful moments, further enriching discussions (Latham, 2003). The role of participant-led photography will be elaborated in the next chapter.

A key advantage of diaries is their ability to capture experiences in real time, reducing retrospective bias and providing richer contextual data compared to traditional methods (Reid et al., 2011). Reflexive diary writing can also serve as a tool for self-reflection and behavioural change, making it particularly useful for sensory ethnography. However, diaries rely on self-reporting, which can introduce biases such as social desirability or incomplete entries due to participant discretion (Reid et al., 2011). Maintaining long-term engagement can also be challenging, though digital diary tools offering short, automated prompts help mitigate this issue (Bolger et al., 2003; Reid et al., 2011).

How does it relate to the sensory modalities: smell and touch?

When studying sensory experiences through diaries, researchers often focus on smell and touch alongside other senses. Participants typically document their overall sensory perceptions with specific prompts. Rodrigues et al. (2021) examined cheese consumption, asking participants to record contextual details along with sensory perceptions and emotions. While 23 sensory characteristics were noted, only two—'strong aroma' and 'mild aroma'—related to smell. Touch-related attributes, primarily

concerning mouthfeel, included ‘greasiness,’ ‘soft,’ ‘harder,’ and ‘thicker crust,’ underscoring its significance in cheese perception. Chen et al. (2012) explored sensory importance in early product experiences using diaries and interviews. Over four weeks of home use, participants tracked sensory aspects of an orange juicer. Touch was most critical, first for exploration and later for evaluating fatigue, comfort, and pain, which heightened stress. Smell, though less emphasized, evoked positive emotions through the scent of fresh juice.

These studies highlight how diaries can capture multisensory experiences. It does seem of importance to specifically ask to describe each sensory modality (or the ones that are to be researched) separately and distinctively. This is due to our highly visually dominant mindset (Pink, 2015), if questions are too open, less dominant or focussed modalities such as smell and touch might not be reported on as much or as profoundly.

Is it used in the workplace to measure design

The diary method is mentioned as a frequently used method to study the experiences close to the time of their occurrence, as well as in real-life setting which cannot be studied in laboratory form (Soucek et al., 2023). Generally in the workplace, it is not always scaled under ethnographic method but also as predominantly quantitative self-report questionnaires (Soucek et al., 2023). This quantitative aspects makes it less ethnographic, but as they mention, there are different types of diary study designs. Soucek (2023) mentions two examples where the diary method was applied in office design (Lindberg et al., 2018; Soucek et al., 2023; Thayer et al., 2010), both mostly focused on visual and acoustic aspects of the office environment, and their relation to employee stress. They both compared participant experiences over multiple consecutive days, in the different environments: private office, high partition cubicle or open bench (Lindberg et al., 2018) and no skylight, no window view vs skylight and window view (Thayer et al., 2010). Beattie and Griffin (2014) too mentioned using the diary method by means of using surveys for over the course of 8 days. To research the effect of experiences of incivility at work on two outcomes, stress and engagement, at a within-person level they used 4 validated questionnaires using Likert scales. So here too, the diary is interpreted as qualitative self-report measurement using questionnaires, opposed to more open ended, qualitative questions. Although, this method does make for a more objective analysis and lower participant burden, it does negate the previously mentioned advantages of a natural, spontaneous and deeper level responses and reflection. Poppleton et al. (2008) do have an example of using qualitative questions in their research in work-life balance among companies, instead of workplace design. Here the diary consists of a short summary of the events and activities of the day, and then a longer section consisting of elaboration questions related to the topic (Poppleton et al., 2008). So there are examples of qualitative use of the diary method, yet it seems to be less frequently used, and often not concerning workplace *design*.

Is it used to measure psycho-physiological stress?

As mentioned above, the chapter of Soucek et al. (2023) mentioned two examples of office design being compared to psycho-physiological stress using the diary method. In one example they explored the association of workplace design and employee health and wellbeing, with physiological stress as heart rate variability and perceived stress levels via smartphone app (Soucek et al., 2023). Another example mentioned, investigated how physical workspace features affect employee stress by room size and windows. Using cardiac monitoring, cortisol sampling, and hourly self-reports, they assessed physiological and psychological stress responses alongside environmental satisfaction (Soucek et al., 2023). These are all quantitative stress measurement (psychological and physiological) methods that are less ethnographic but are well-researched and validated. But there are also diary methods measuring stress that are more qualitative. Clarkson and Hodgkinson (1999) the quantitative application focus, and aimed to explore a more 'free response format', qualitative style to complement this quantitative measurement. In this free format, participants were presented with a diary booklet in which they were requested to record self-identified critical incidents that had triggered a perception of stress, their feelings and their perceived consequences. The format was free but the daily explanation did include prompt questions as: *'What was the source of the stress', 'What was the consequence of that stress (how did it make you think, feel behave. Did this have effect on you personally/the organisation?)' and 'How did you cope'* (Clarkson & Hodgkinson, 1999). They found this method to be a useful method to complement traditional questionnaires to gain deeper insights in the experiences of occupational stress (Clarkson & Hodgkinson, 1999). Mazzola et al. (2011) also stress the importance of qualitative methods in addition to quantitative measures, especially in work-related stress research, as there can be any variety of stressors, the provision of context and gaining deeper understanding is key. They too discuss examples of more qualitative ways of performing diary research, but in a bit less free format. They describe the method to include open ended survey questions which are answered at certain times of the day, where the participant experiences a stressful event. There seems to be a range on how free the diary method can be applied to measure psychological stress, each with its own pro's and con's, where at least the addition of gaining qualitative data has been identified as very valuable.

2.3.6. Participant-led-photography

Participant-led photography, also known as visual diaries or photo-voice, is a qualitative research method where participants take photographs to document their experiences, surroundings, and daily lives (Burles & Thomas, 2014; Pilcher et al., 2016). Rooted in participatory research, this method allows participants to actively contribute to data collection, enhancing its authenticity (Pilcher et al., 2016). Typically, participants receive instructions related to the research topic and take photos accordingly. These images are then used in interviews or focus groups to facilitate discussions and elicit deeper insights into participants' memories, emotions, and meanings attached to the photos (Burles & Thomas,

2014; Latham, 2003). This combination of photography and interviews helps capture narratives closer to the actual experiences, improving accuracy (Latham, 2003). While most studies use participant-led photography alongside interviews, some quantitative approaches exist, such as analysing uploaded images with digital tools (Bakolis et al., 2018). Another emerging method involves using Instagram for research, either as a digital life archive or a platform for storytelling and data sharing (Shortt & Warren, 2020).

The main advantage of this method is its ability to capture habitual aspects of daily life that might go unnoticed in traditional research (Pilcher et al., 2016). It enables participants to reflect on and articulate experiences beyond verbal communication, making research more engaging and insightful (Burles & Thomas, 2014). However, limitations exist—visual diaries may reflect a selective self-representation rather than a full depiction of daily life. Additionally, translating abstract or emotional experiences into photographs can be challenging, particularly when dealing with sensitive topics (Burles & Thomas, 2014). Moreover, not all aspects of daily life are visible or easily photographed, limiting the method's ability to capture intangible experiences (Burles & Thomas, 2014).

How does it relate to the sensory modalities: smell and touch?

Photography is inherently a visual concept, and is therefore mostly linked to visual research. Yet, this does not mean that it cannot play a valuable role in the research of other sensory modalities. The sense of smell, or olfaction, is notoriously difficult to document and study because it lacks a direct visual component. However, participant-led photography could still play a role in capturing the contexts and sources of olfactory experiences. Participants can photograph places, objects, and activities associated with particular smells, thereby providing visual proxies for olfactory data (Pink, 2011). For instance, a participant might document the kitchen where they prepare food, a garden with fragrant flowers, or a marketplace with diverse aromas. These images could help recover memories (Latham, 2003), and combined with participants' narratives about the smells they experience, could offer valuable insights into the cultural and personal significance of olfaction.

By incorporating the sense of touch, researchers can explore how tactile experiences, along with other sensory modalities, are intertwined with visual ones (Pink, 2011). Pictures taken during the research can depict materiality, patterns and textures of the participants surrounding (Shortt & Warren, 2020). For example, participants might photograph their hands touching different materials, capturing the sensory qualities of these interactions through embodied touch. These images can then be analysed to understand the role of touch in participants' daily lives and how it contributes to their overall sensory experience. This method aligns with Pink's argument that visual and sensory ethnography must account for the multisensory nature of environments and experiences, recognizing that images can evoke and represent non-visual senses through associative and contextual means (Pink, 2011). Although, these arguments show that there could be potential to use participant-led photography, in combination with

interviews and/or diaries, as a method to research touch and smell, there seem to be little to no cases where this has explicitly been done.

Is it used in the workplace to measure design

The use of participant-led photography in workplace design research remains rather limited, but not entirely uncommon. Warren has explored its application in several studies (Shortt & Warren, 2019, 2020; Warren, 2002, 2008). She found that words alone were insufficient to describe workplace environments, leading her to ask participants to take Pictures with the briefing: *'Show me what it feels like to work here'*, which were then discussed and contextualised in interviews (Warren, 2002). She argued not to over-emphasize the importance of the visual as these images often conveyed underlying emotions or rituals. For example a photograph of cookie bags symbolizing a sense of community. This notion under-scribes the importance of interviews and notes alongside the photography. A later study used the same method to evaluate a redesigned office space, reinforcing the value of combining photography with interviews (Warren, 2008). This combination of participant-led photography in combination with interviews was also performed by Pataki-Bitto (2024), where she asked employees with a hybrid work schedule to photograph office elements that inspired their work or boosted well-being, followed by interviews to discuss the meaning behind their photos. Through this qualitative method she was able to uncover that the social, cultural and shared memory influences in the workplace environment to have a big influence on the employee perceived wellbeing (Pataki-Bitto, 2024). The sensory aspects was not focused on.

Despite critiques regarding analysis, Shortt and Warren introduced Grounded Visual Pattern Analysis (GVPA) to systematically interpret visual data, integrating dialogical and archaeological analysis (Shortt & Warren, 2019). They demonstrated its use in an ethnographic study of hair salon workers, uncovering insights into workplace environments and identity formation. Herneoja et al. (2023) also applied a less photography method that was then analysed by systematic models. They analysed the participants gaze and view from their workstation, and asked participants to take pictures through a smartphone application, attaching location and time stamps (Herneoja et al., 2023). These more systematically and technically analysed methods might help in removing subjectivity and help recognising patterns, but it lacks in the ability for participants to attach meaning and context to the pictures and design elements, which is essentially the key benefit of ethnography. Although research on participant-led photography in office design remains fairly limited, there is some work that highlights its potential beyond visual aspects, emphasizing its effectiveness when combined with interviews.

Is it used to measure psycho-physiological stress?

Participant-led photography is often used to explore general experiences rather than focusing on a single concept like stress (Burles & Thomas, 2014; Latham, 2003; Pilcher et al., 2016; Warren, 2002). While stress may emerge as a theme, most studies use open-ended prompts, allowing for a broader, more

holistic perspective (Glaw et al., 2017). However, this approach can make specific measurements more challenging.

Some studies have applied participant-led photography to examine distress more directly. For example, Kantrowitz-Gordon and Vandermause (2016) investigated distress among new parents of premature babies by asking them to take digital photos over 2–4 weeks that represented their feelings after bringing their baby home. These images were later discussed in interviews, helping parents articulate their experiences and making distress more tangible through metaphor (Kantrowitz-Gordon & Vandermause, 2016). Other research has combined participant-led photography with quantitative psychological stress measures. Jiang et al. (2019) studied sweatshop workers' perceptions of outdoor environments by asking them to photograph spaces that influenced their mental state. Participants then selected three key photos, reflected on them for a minute, and completed a stress and anxiety questionnaire before discussing their emotions and memories (Jiang et al., 2019). Similarly, Girang et al. (2022) used participant-led photography and walk-along interviews to explore stress-alleviating spaces on a university campus. Their focus was on social settings, distinguishing between high- and low-energy environments and solitude versus social interaction. By combining photography, mapping, and interviews, they identified which environments were sought out in different emotional and stress-related situations (Girang et al., 2022).

These studies demonstrate that participant-led photography can be valuable for investigating stress, particularly when combined with interviews or quantitative measures. However, its strength lies in capturing the complexity of experiences rather than isolating single variables.

2.3.7 Sensory mapping

Sensory mapping depicts the world as it is subjectively experienced through various human senses, combining aesthetics with empirical data to highlight personal, time-specific realities and the more-than-visual characteristics of places. Contemporary practitioners use diverse sensory media to emphasize the emotional, embodied, and affective dimensions of cartography (McLean, 2020b). Most of the sensory maps used, are single-sense maps, focussing on one of the five main senses yet there are also attempts of performing the method multi-sensory (Misharina et al., 2023). The representation of the sense, however, does not always have to be in the medium of original encounter (McLean, 2020b). The medium of the map is often visual even though the researched sense might not be. Sensory mapping often explores how sensory stimuli influence human perception and experience of space. It investigates the qualitative aspects of sensory experiences and how these contribute to the understanding and attachment to specific places (Misharina et al., 2023).

Sensory mapping is typically performed by providing participants with a base map of the area to be studied, which they use to record their sensory experiences (Misharina et al., 2023). Participants are encouraged to note down impressions such as specific smells, sounds, and tactile sensations at various

locations. These impressions are often marked using color-coded stickers on the map, which represent different types of sensory stimuli (Misharina et al., 2023). The mapping process quite often include wandering through the area, allowing sensory stimuli to guide the participants' paths and experiences (McLean, 2020b), very comparable to the sensory walks mentioned in the section of observations (Pink, 2015). Data collected from individual maps are then digitized and analysed collectively to identify patterns and create a comprehensive sensory map of the area (Misharina et al., 2023). An example of method was proposed by Degen et al. (2016) from a workshop they gave, using sensory mapping through a Neighbourhood in London as example. They proposed creating three maps. The first map captures sensory experiences: smells, sights, textures, and sounds. The second map records subjective experiences from interviews, detailing feelings about the space as hostile, friendly, oppressive, etc. The third map, a 'relationship map,' asks users why they perceive the space in those ways. The goal is to merge the researcher's observations with the users' experiences to create 'places of translation' (Degen et al., 2016). It is important to question what the added advantage of the mapping would be to the sensory walks in itself. It does create a more visual output, but it might also aid in the reflection and recollection process of the sensory experiences, rather similarly as participant-led-photography does.

How does it relate to the sensory modalities: smell and touch?

Sensory mapping is a multisensory methodology that explores all five senses, with smell being the most commonly studied, followed by sound and vision (McLean, 2020b). Smell maps are often created through smell walks, where participants navigate an environment, identifying and documenting smells. Sometimes, they are blindfolded to enhance perception. The recorded data—locations, sources, and characteristics of smells—are then visualized in various formats, including Scratch-n-sniff walls, 3D tactile maps, and traditional 2D visual maps (McLean, 2020b). Smell mapping helps localize odours, understand their environmental interactions (e.g., wind, humidity), and document their cultural significance. McLean's (2012) city smell maps (e.g., Edinburgh, Amsterdam) use modified cartographic symbols to depict scent dispersion, while others use dots to highlight humidity-driven scent landscapes.

Touch, however, is rarely studied in sensory mapping. Misharina et al. (2023) found that participants often overlooked touch in a city-based multisensory study. They found that people perceived two categories of touch: 1) being touched by an object, human or weather, and 2) the experience of intentionally reaching to touch something. The last category, the conscious choice, often came with the revelation that the experienced tactile properties often did not match the visually perceived and expected tactile properties (Misharina et al., 2023). From this example we can learn that researching touch in a mapping method is certainly possible, but it is important to guide the participants well in including touch in their conscious experience. This suggests that studying touch through mapping requires

prompting participants to be more consciously aware of their tactile experiences. Touch is more commonly integrated as an output in tactile mapping rather than as an input for analysis (McLean, 2020b).

Is it used in the workplace to measure design

Sensory mapping is often performed in an urban scale involving a city or neighbourhood setting (Degen et al., 2016; McLean, 2020b; Misharina et al., 2023), letting participants walk through a neighbourhood, and map out their experiences on a map. One might consider whether a similar method could be applied in an office environment using an office layout map. Given the smaller scale and distances, the predominantly indoor environment, and the likely uniform design theme throughout the building, the observed differences may be less pronounced than in an urban setting. However, this does not diminish the method's potential usefulness. Variations likely exist within the office layout, with certain areas being more popular for specific activities and providing distinct sensory inputs. These differences could still be effectively mapped on an office layout plan. There is an example of an indoor setting where a smellscape sensory map has been performed in a hospital corridor (McLean, 2020a). This project investigated how hospitals can be understood through their unique smellscape, formed by interactions of people, cleaning equipment, and food (McLean, 2020a). A virtual map, requiring imaginative engagement, captures smells from weak coffee near the canteen to metallic blood in operating theatres. Research involved gathering simplified smell descriptions from open-ended questions posed to friends and colleagues, with artistic interpretations of these smells presented as "smell shapes." The smells were colour coded and visualised creatively on a map of the corridors (see figure 1, (McLean, 2020a)). These indoor settings are very rare for sensory mapping, but this example does show that it is certainly possible and possibly quite helpful. Although, this is not an office setting, but it is a workplace. The application of the method would not be too different.

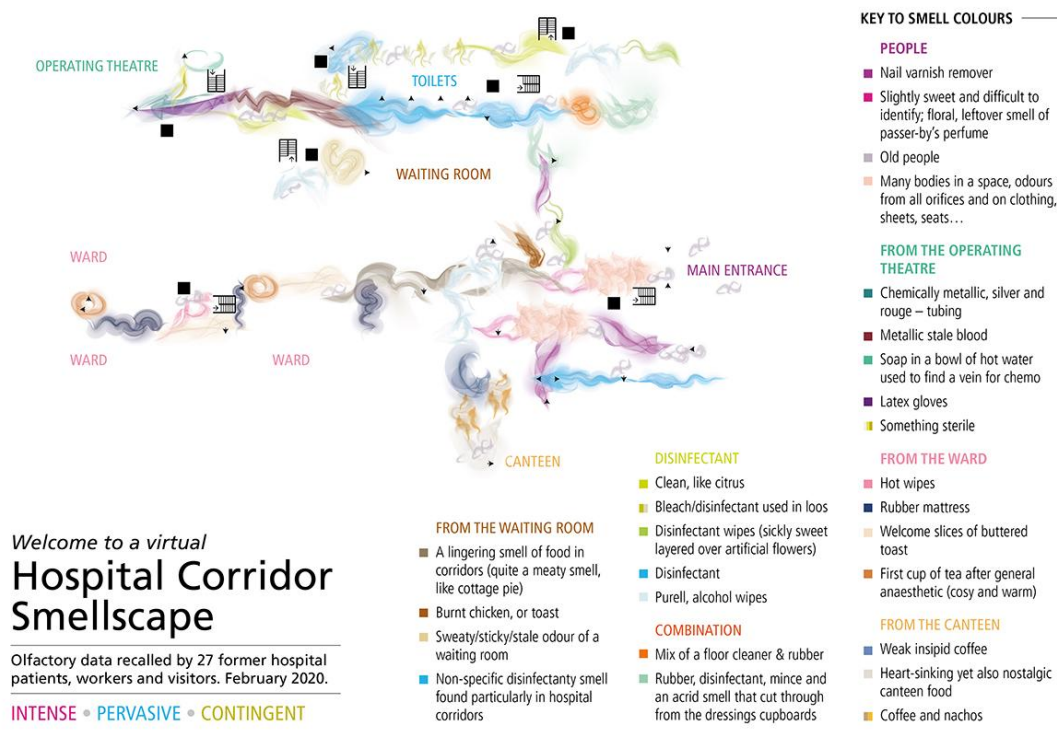


Figure 1: An example of what sensory mapping for smells could look like in an indoor setting. This example (McLean, 2020a) performed a sensory mapping study to investigate the smell experience in hospital corridors.

Is it used to measure psycho-physiological stress?

Sensory mapping is rarely used to measure stress specifically. Measurements that come closest are measures of emotion. For example Hogertz (2010) measured the arousal dimensions of emotions of high-low arousal against positive-negative, where stress is in the negative arousing quadrant (Hogertz, 2010). They used real-time electrodermal activity (EDA) as an indicator of emotional brain processing, which was then compared with arousal-eliciting environments on the path the participant walked and projected on an emotional arousal map. This is an example of physiological measurement, but most mapping research is done with psychological measurements, where people comment on their emotions during the walking a qualitative form (McLean, 2012; McLean, 2020a; Misharina et al., 2023). For example in a study using a scratch and sniff map, participants were asked to freely add sticky notes with places or feelings the smell reminded them of (McLean, 2012). This way it is difficult to capture stress, or any specific emotion, but more useful for a more explorative method.

Rationale for the empirical study

Now that the literature has shed some light on the existing body of research, both on sensory effects on psycho-physiological stress in the workplace environment, as on sensory ethnographic techniques, the next step is to answer the sub-question: What can we learn about the relationship between sensory ethnographic techniques found in the literature review and psycho-physiological stress resulting from the physical workplace environment? To answer this question a pilot study was conducted.

From the literature was found that there was a body of work of visual and auditory influences of the workplace environment on stress, yet the sensory dimensions of smell and touch were largely neglected. This pilot study will therefor focus on these two modalities. From the literature we also found that traditional quantitative measurements often lack depth and context insights to evaluate the sensory experiences and effects (Gobo, 2011; McGibbon et al., 2010; Pink, 2013), whereas sensory ethnography is sensitive to being very subjective (Pink, 2015). This pilot study will therefore combine more traditional quantitative methodologies for measuring psychological as well as physiological stress with sensory ethnographic measurements to answer the sub question: What can we learn about the relationship between sensory ethnographic techniques found in the literature review and psycho-physiological stress resulting from the physical workplace environment?

To do so, this pilot set up an experiment where we expose participants to 4 different environments. In these rooms the sensory modalities of touch and smell will be manipulated, as the literature study found that those senses have been under researched, but could provide interesting results (Pallasmaa, 1994; Spence, 2020a). Their experiences will be evaluated both by sensory ethnographic techniques of focus groups and sensory diaries. The focus groups were chosen as from literature was found that many sensory ethnographic techniques are combined with focus groups or interviews, as they seem to provide much context and understanding in underlying mechanisms and reasoning (Bolger et al., 2003; Latham, 2003; Low, 2015; McLean, 2020a; Pink, 2015). The diary studies were described as an effective means to gain in situ information (Bolger et al., 2003; Latham, 2003), especially ass the focus-groups are not possible to be held in situ.

By exposing participants to different environments within a single day, the study aimed to simulate the variety of sensory experiences that might be encountered over a week, while keeping the data collection manageable. This approach also allowed for a comparison of the subjective experience of the environments and psychological stress, and the physiological stress responses to understand how both sensory input and personal perception affect stress levels.

3 METHODS

3.1 Study design

This experiment employed a 2 (materiality: hard vs. soft) x 2 (scent (odourless vs. citrus) within-subject design. Participants worked in four environments over the course of a single workday. Participants' responses were measured using a combination of quantitative and qualitative methods across three types of data collection: objective physiological stress measurement (EDA, HRV), self-reported stress (EMA), Sensory Diary, and Focus groups.

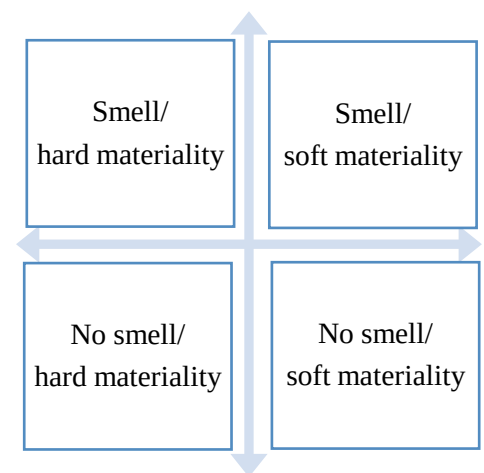


Figure 2: graphical presentation of the two-by two study design room conditions.

3.2 Participants

Participants were a convenience sample from the students and employee population of Eindhoven University of Technology (TU/e) invited via the JF Schouten participant database and direct on-campus recruitment. All participants received a compensation of 30 euros for their time and involvement in the study. A total of 20 to 30 participants were deemed necessary to ensure sufficient data collection and variability, yet due to monetary and time-restrictions, only 14 participants were recruited from which 3 dropped out due to personal and scheduling issues resulting in 11 participants. Although a mix of students and employees would have been preferred, the entire participant group was made up of TU/e students (5 women, 6 men), due to the time consuming study.

To be eligible for the study, participants had to meet the following criteria (Beil & Hanes, 2013):

- Be at least 18 years old.
- Be able to read and speak English.
- Have no heart conditions, or epilepsy.
- Not be taking medications such as antidepressants or beta-blockers (although participants taking oral contraceptives or light painkillers, such as ibuprofen, were allowed to participate).
- Not regularly experiencing panic or anxiety that might be triggered by stress.

Since the study involved reflecting on sensory experiences, individuals with conditions that could impair their ability to perceive sensory input were excluded. This included participants with:

- Impaired sense of smell (e.g., Anosmia. Also including temporary conditions, such as due to the flu or COVID-19).
- Impaired or decreased haptic (touch) perception.
- Uncorrected visual impairments, such as colour blindness.

3.3 Procedure

Participants completed the study over the course of a single day, experiencing four different preselected environments in a within-subject design. Each day a maximum of four participants was possible, the complete week schedule can be found in appendix B. Figure 3 shows the procedure, followed by a step by step explanation.

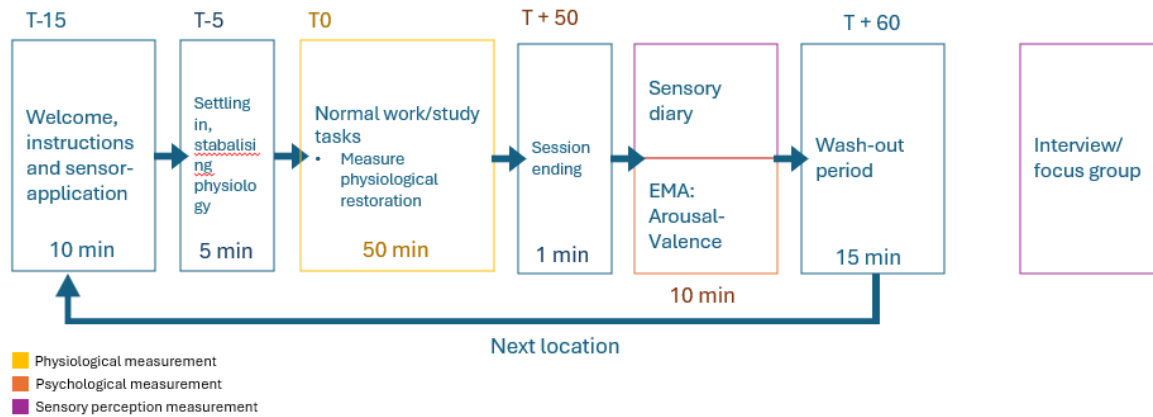


Figure 3: a schematic overview of the procedure steps, and the exposure and measurement times. Each participant would walk through these steps for all 4 of the conditions.

T-15 At 09:00 welcome, explanation, informed consent and sensor application. Upon arrival, participants were greeted and introduced to the study giving a brief overview of the day's activities, ensuring they understood the purpose of the research and their role. After reviewing and signing the informed consent form, they were connected to physiological monitoring equipment.

T-5 Acclimation period. The participants were getting settled into the room and take out their stuff. These 5 minutes of data are not used in the analysis.

T0 Work or study tasks. In each environment, participants engaged in a 50-minute work session. The tasks were partially controlled to ensure consistency across participants, with activities focused on quiet, individual work such as reading or writing. Participants were asked to avoid phone calls, meetings, moving around, eating or any other highly engaging activities during this time to prevent confounding effects on stress levels.

T+50 EMA questionnaire and sensory diary. The participants were given 10-15 minutes to consider their environment and complete a sensory diary and EMA questionnaire.

T+60 Washout period. between environments, participants underwent a 15-minute washout period or break to minimize any lingering effects from the previous setting. During this time, participants were asked to not engage in any heartrate altering activities such as physical activity.

The steps above were repeated for all four environments, with participants moving from one sensory condition to the next in a randomized order. Each participant was exposed to both soft and hard textures, as well as smell and no-smell environments. The work tasks remained consistent across the conditions to ensure comparability.

End-of-Day Focus Group. At the end of the day, participants took part in a focus group to discuss their overall experiences in each environment and reflect on their stress responses. Photograph probes were used to prompt discussions about the sensory aspects of each environment. Prior to the focus group, the researcher conducted a quick preliminary analysis of the sensory diaries and psychological stress data to tailor the discussion and ask more directed questions based on initial findings.

By the end of the day, each participant had experienced all four environments, allowing for a comprehensive comparison of both objective and subjective stress responses across different sensory conditions. The precise time-schedule can be found in appendix A.

3.4 Study environment and exposure

Location

The study took place in the Atlas building of Eindhoven University of Technology (TU/e), using two designated spaces: General Purpose Lab 1 and the Gameplay Lab. Both labs were used to simulate the four different environmental conditions by varying the furnishings while keeping the same physical room layout (See figure 4 and 5). All rooms featured large windows to provide a consistent natural light source across conditions. Each participant experienced all four environments for an hour over the course of a single day, with each environment's sensory manipulation carefully controlled.

Four distinct environments were created by combining the presence or absence of a pleasant citrusy smell with soft or hard material textures.

Smell Dimensions

In this study, participants were exposed to either a citrusy aroma or no smell. To ensure consistency, aroma oils were used to diffuse the smell throughout the room, as these oils are often rated as pleasant and can maintain a relatively stable concentration over time (Herz, 2009; Spence, 2020b). The scent was distributed by a stick-diffuser from H&M winter spices, including citrus and ginger notes as well as warm spices such as cloves, almonds and vanilla which are generally considered pleasant and are congruent with the time of year (Herz, 2009; Spence, 2020b).

Texture Dimensions

The texture manipulation involved staging rooms with either soft or hard materials to influence the tactile sensory experience. The "soft" rooms were furnished with items such as couches, poufs, blankets,

pillows, tablecloths, and curtains, while the "hard" rooms were outfitted with hard tables, plastic chairs, and other rigid materials. The orientation, monitor set-up and chairs were the same in all four rooms. To confirm the validity of these manipulations, the thesis supervisors evaluated each room after staging to ensure that the furniture and materials accurately represented "soft" or "hard" environments.



Figure 4: Soft room furnishing. Soft elements include a couch, stuffed animals, pillows, soft chair, soft material table, plants, a table cloth and a blanket over the desk chair.



Figure 5: Hard room furnishing. Hard elements include a hard materiality table with chairs. There are no extra soft elements such as blankets etc over the chair. The desk chair itself does have a rather soft surface, as these desk chairs are important for good posture and wellbeing of the participants during the study.

3.5 Data collection

We employed a combination of quantitative and qualitative methods across four types of data collection: objective physiological Stress measurement (EDA, HRV), self-reported stress (EMA), sensory diary, and focus groups.

3.5.1 Psycho-physiological Stress Methods

Physiological Stress Measurements

To monitor physiological stress, two key biometric indicators are used: heart rate variability (HRV) and electrodermal activity (EDA) (Yin et al., 2020). Both are measured using the Mobi-8 device, recording

to via the desktop app PhyLo. The HRV electrodes are placed using the 3 electrode system as per device instructions (left clavicle, right clavicle, left underneath ribcage). It collected raw data from a single-channel electrocardiogram (ECG), which was then used to derive secondary parameters such as HRV. The sampling rate was set at 1024 Hz. For HRV, we calculated a time-domain indicator: the root mean square of successive differences between normal heartbeats (RMSSD, in milliseconds [ms]). A higher RMSSD value reflects increased parasympathetic activity, which implies stress reduction.

EDA measures the electrical conductance of the skin, which increases with sweat gland activity driven by sympathetic nervous system arousal, measuring the galvanic skin response (GSR). It was measured via finger electrodes, placed on the fingertips of the ring and middle finger of the non-dominant hand, using Velcro wrapping, as per device instructions. Due to lacking time, and noisy data caused by usage of the hands during the deskwork, this EDA data has not been processed and used.

To minimise disturbances in the physiological data, the participants were instructed to remain seated, not eat during the measurement time, refrain from drinking caffeinated drinks that day, not excessively drink alcohol from the night before or perform (heavy) physical activity that day

Psychological Stress Measurements

Momentary affect was assessed as proxy for psychological stress, using a questionnaire measuring the three-dimension core affect model by Schimmack and Grobb (2000) including the dimensions: Pleasure-Arousal-Tension (PAT). Each dimension was represented by three bipolar indicators, calculated by subtracting opposing affective states. Participants rated 18 mood items on a 4-point scale (1 = I do not feel it, 2 = I feel it slightly, 3 = I feel it moderately, 4 = I feel it strongly), following a standardized instruction. The items for each dimension were: pleasant-unpleasant, positive negative, and good-bad for pleasure (Cronbach's $\alpha = 0.7$); awake-sleepy, alert-tired, and wakeful-drowsy for awake (Cronbach's $\alpha = 0.9$); and restless-at rest, tense-relaxed, jittery-calm for tension (Cronbach's $\alpha = 0.9$) (Schimmack & Grob, 2000).

Participants completed the questionnaire in each of the rooms after the working session while seated in the study environment, using the m-Path app on their smartphones. The complete questionnaire can be found in appendix C.

3.5.2 Sensory Ethnographic Methods

Sensory Diary

A semi-structured diary method was employed to capture participants' subjective sensory experiences in each environment. This diary combined open-ended questions with Likert scale ratings, allowing for both rich qualitative and quantitative data collection (Bolger et al., 2003; Latham, 2003; Pink, 2015). By directing participants' attention to specific sensory modalities—particularly smell and touch, which

are often less consciously perceived—this method encouraged more focused reflection on the sensory aspects of each environment, while still being present in the environment for optimal recollection as they are still in the environment experiencing the senses (Pink, 2015). The diary consisted of questions organized into sections for each of the four sensory modalities: vision, sound, smell, and touch. Additionally, general questions about the overall sensory experience, frequency, and intensity of stimuli were included. The questions about vision and auditory experiences were asked as well, to help avoid skewing participants' attention away from the primary sensory dimensions being studied (smell and touch) and still provide a holistic overview of their sensory experiences (Pink, 2015; Spence, 2020a). The sensory appraisal questions were formulated as follows: 'How would you rate the smell in the room?' With a follow-up open ended question asking them to explain the rating.

Participants were asked to fill out the sensory diary digitally on their smartphones immediately after completing the ecological momentary assessment (EMA) questionnaire in each environment, using the same software: m-Path. Digital diaries were chosen based on recommendations from Bolger (2003), who highlighted that digital methods enhance data accuracy by allowing real-time time-stamping, reducing participant forgetfulness, increasing reflection, and minimizing errors in data entry. The complete diary questionnaire, including all questions and scales used, can be found in Appendix D.

Focus group

At the end of the study day, participants engaged in a focus group to discuss their experiences in the four different environments. The focus group was conducted on the same day to ensure that participants' memories of the environments were still fresh. The amount of participants per group was dependent on how many participants signed up for the day, ranging from one (interview) to four. During the day, the researcher performed a preliminary analysis of the diary and stress data on the first three rooms. This quick review looked at the diary and EMA results of the first three rooms. The (large) noticeable differences in EMA and quantitative sensory ratings were compared per room manually, and aligned with the open question answers. This helped form more directed and relevant questions during the focus group discussion, enriching the data collection with deeper insights.

The focus group followed a structured approach based on Hennink et al. (2020). The session unfolded in the following stages: Introduction and explanation, introductory warm-up questions, key questions (focused on smell and touch experiences, and based on diary entries), closing question and conclusion and thanking of the participants. The complete focus groups guide can be found in appendix E. The focus group ran until saturation, but generally lasted between 20 – 30 minutes. The discussions were audio-recorded via MS Teams as well as using the audio recorder on the researchers smartphone.

Since the focus group could not be conducted in the environments themselves, photographs of each room were used as memory probes. This visual aid helped participants recall specific sensory details and experiences from each environment, facilitating a more detailed and accurate discussion (Latham,

2003; Pink, 2015). By using these probes, the focus group method compensated for the change in physical setting while still capturing the richness of participants' sensory experiences. The entire focus group scheme can be found in appendix E.

3.6 Data analysis

The data analysis will be built up in two main sections, qualitative analysis and quantitative analysis. The quantitative analysis will explore the relations between room manipulations (smell and soft materiality) and psycho-physiological stress. From the literature we learned that the personal preference might mediate the effect the sensory experience has on the psychophysiological stress, therefore the quantitative parts of the sensory diary will be analysed here as well. Figure 6 shows a representation of a mediation analysis. The quantitative section will first analyse the separate paths. It will analyse relations between the room conditions (smell and softness) and psychological stress and physiological stress (path c), then similar analyses on the relation between the room conditions and the sensory appraisal (path a), and then a mixed analysis on the effects of sensory appraisals on psychological stress and physiological stress (path b). If all these analyses are significant, a mediation analysis will be performed. If one of these paths is not significant, there would be no indication of a mediation effect and would therefore not be performed. All statistical analyses are performed in Stata BE 18 (64Bit).

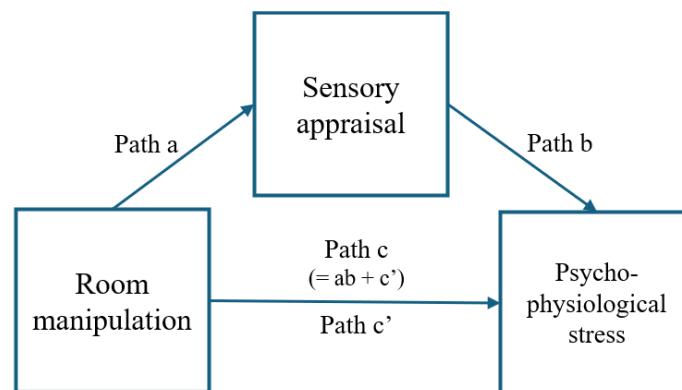


Figure 6: A visual representation of the possible mediation effects that the room manipulations could have on psycho-physiological stress, with possible mediation by the sensory appraisal.

The qualitative analysis will consist of a thematic analysis concerning the open questions of the sensory diaries and the focus groups. To remain as unbiased as possible, the different data types were first analysed separately before being compared.

3.6.1 Psychological stress

To analyse the effect room conditions have on psychological stress, a two-way repeated Anova's have been performed. These models have been tested for normality of the residuals using a Shapiro-Wilk test. The sphericity was not tested for as there is a degree of freedom = 1. When the normality

assumption was not met, transformations were tried. If they did not improve the normality, a Mixed model would have been performed, as they are more robust.

The psychological stress was also analysed with sensory appraisal as predictor. For this a Linear Mixed Model was performed, accounting for repeated measurements using participant ID as random intercept. These models were tested for the normality of the residuals assumption using the Shapiro-Wilk test, homoskedasticity using visual inspection of a residual vs fitted scatterplot.

3.6.2 Physiology

The raw heart rate variability (HRV) data was downloaded from PhyLo (after being measured with the Mobi 8 device) and analysed using Kubios HRV Scientific 4.1.1, a validated software for HRV processing. Artefacts and noise in RR intervals were automatically corrected and manually checked based on established guidelines, with noise detection set to medium, and the beat correction on the standard 0.3 (Lipponen & Tarvainen, 2019). Detected ectopic beats and abnormal RR intervals were corrected through interpolation, ensuring data integrity (Lipponen & Tarvainen, 2019). Noise segments were disregarded from the analysis. Samples with more than 5% corrected beats or more than 5% noise were excluded. This has led to the exclusion of 1 participant data, (for all rooms, all samples) and one extra sample. RMSSD values falling outside the healthy range (6.9–99.4 ms; Agelink et al., 1998) were also excluded, which has led to the exclusion of three rooms for 1 participant, and four other samples. HRV was analysed in three five-minute time bins (5–10 min, 25–30 min, 45–50 min). Mean RMSSD per condition was compared to all three of the psychological stress dimensions by means of linear mixed model. Furthermore, a mean RMSSD effect was analysed as dependent variable for effects by the predictor room manipulations by means of a repeated Anova. This model is also tested on the assumptions of normality of. A linear mixed model was run to investigate the effect of sensory appraisal on the mean RMSSD, with ID as random intercept to account for the repeated measurements. To check the RMSSD progression throughout the working hour, the three time bins will be assessed visually, for possible stress restoration or progression.

3.6.3 Diary

The diary entries are separated into two sets of data: the quantitative data containing the answers from the sensory appraisal on a Likert-scale questions, and the qualitative, open-ended questions.

For the quantitative data, the sensory appraisal was used as independent variable in the LMM to assess on both psychological stress and physiological stress, as described in the sections above. It was also analysed as dependent variable in the analysis studying the effects of the room conditions (smell and soft), in a two-way repeated Anova. This model was checked for the assumptions linearity of the residuals using a Shapiro-Wilk test, the independence of observations through independent measurement design. The sphericity was not tested for as there is a $df = 1$. When the normality

assumption was not met, transformations were tried. If they did not improve the normality, a Mixed model would have been performed, as they are more robust.

The open-ended responses from the diary entries were thematically coded using the same coding scheme as the interviews, which will be elaborated on more in the next section of the focus groups.

3.6.4 Focus groups/ interviews

The data obtained from the focus groups was analysed using thematic analysis. First, the audio recordings were transcribed verbatim using transcribing software Alice and checked manually. Subsequently, an initial set of codes was developed based on the research question and recurring patterns or terms from the data to capture participants' experiences. The transcripts were then systematically coded, allowing for the emergence of new codes when necessary. At the end, the transcripts were revisited to ensure that all relevant data was coded appropriately with the newly emerged codes included.

To gain overall insights, a matrix was created to examine the frequency and distribution of codes across the dataset setting out the codes per transcript. Additionally, relationships between codes were explored in a interaction matrix to identify patterns and connections. For the theme development step the codes were organised into meaningful groups, revealing patterns and relationships that provide insights into the research question. This process moves from detailed code analysis to a more abstract interpretation, where themes emerge. These themes are more than recurring elements—they represent deeper, patterned meanings that connect the research questions to the data.

3.7 Ethics

The study followed the ethical principles on psychological research as formulated by the TU/e and the Helsinki declaration. And was approved by the university's ethical board (#HTI 2139). Participants provided informed consent and received monetary compensation of €30 for their time.

4 RESULTS

This results section is built up in two parts: Quantitative analysis and qualitative analysis. As per the mediation analysis figure 6 (chapter 3.6) in the data analysis section of the method, the quantitative section will first analyse the separate paths. Starting with the effects of the room manipulations (smell and softness) and psychological and physiological stress (path c), then similar analyses on the effects of the room conditions on the sensory appraisal (path a), and then a mixed analysis on the effect of sensory appraisal on psychological stress and physiological stress (path b). If all these analyses are significant, a mediation analysis will be performed. The qualitative section will show a thematic analysis on the open questions from the diary, and the focus groups.

4.1 Quantitative Analysis

4.1.1 Room condition and psycho-physiological stress analysis

Psychological stress

Tension

To gain insights in the estimated mean tension values per each condition, these have been displayed in Figure 7, as well as the individual participant values. The lowest tension is reported in the Smell/hard room condition (room 1), whereas the highest tension is reported in the soft/no smell condition (room 4), but the differences in the means seem minimal. Higher tension values present higher reported feelings of tension. The individual values underscore the high variance in tension values.

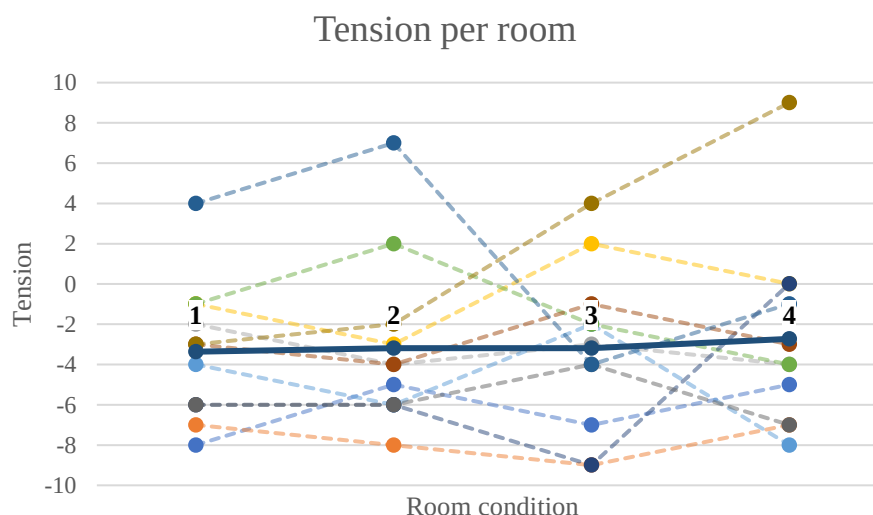


Figure 7: The mean and individual values of the tension dimension of psychological stress. The dotted lines represent individual values, whereas the solid line shows the mean. The room conditions: 1(smell/hard), 2 (no smell/hard), 3 (smell/soft), 4 (no smell/soft)

To test differences between these conditions (smell yes vs no and soft vs hard), a repeated two-way Anova was conducted with Tension as dependent variable and Soft and Smell as factors. The residuals followed a normal distribution (Swilk- test $p = 0.22$). The model showed no significant effects. It

reported for smell $F(1,10) = 0.16$ ($p = 0.70$), for soft $F(1,10) = 0.07$ ($p = 0.80$) and for the interaction of smell and soft partial $F(1,10) = 0.05$ ($p = 0.82$). This means that this data did not find any evidence for an effect of the room conditions soft or smell on tension.

Pleasure

To gain insights in the estimated mean pleasure values per condition, these have been displayed in figure 8, as well as the individual participant values. The highest pleasure was found in the soft/smell condition (room 3) whereas the lowest pleasure was found in the soft/no smell (room 4) condition.

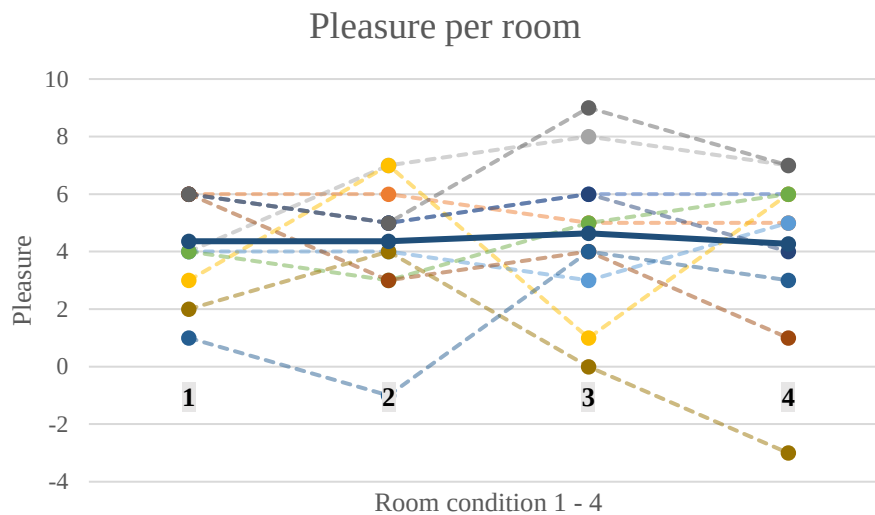


Figure 8: The mean and individual values of the pleasure dimension of psychological stress. The dotted lines represent individual values, whereas the solid line shows the mean. The room conditions: 1(smell/hard), 2 (no smell/hard), 3 (smell/soft), 4 (no smell/soft)

To test for an effect of the room conditions (smell and soft) on pleasure, a repeated two-way Anova was conducted with pleasure as dependent variable and smell and soft as factors. The residuals followed a normal distribution (Swilk-test $p = 0.91$). The model showed no significant results, providing for smell $F(1,10) = 0.10$ ($p = 0.76$), for soft $F(1,10) = 0.02$ ($p = 0.90$) and for the interaction between smell and soft $F = 0.28$ ($p = 0.61$). This means that this data did not find evidence for an effect of the soft or smell condition on the pleasure dimension.

Awake

To gain insights in the estimated mean awake values per each condition, these, as well as the individual participant values, have been displayed in figure 9. The highest awake rating was found in hard/no smell condition (room 2), the lowest was found in the soft/smell condition (room 3). Again, the figure shows a high variance among participants.

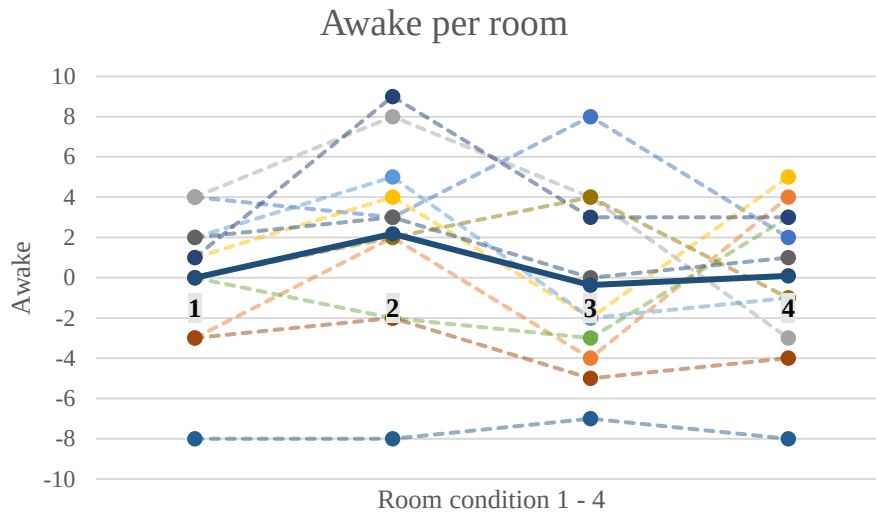


Figure 9: The mean and individual values of the awake dimension of psychological stress. The dotted lines represent individual values, whereas the solid line shows the mean. The room conditions: 1(smell/hard), 2 (no smell/hard), 3 (smell/soft), 4 (no smell/soft).

To test for an effect of these conditions (smell and soft) on the awake dimension, a repeated two-way Anova was conducted. The residuals followed a normal distribution (Swilk-test $p = 0.10$). The model found no statistically significant results. It reported for smell $F = 2.10$ ($p = 0.18$), for soft $F = 2.98$ ($p = 0.12$) and for the interaction between smell and soft $F = 1.02$ ($p = 0.34$). This means that this data did not find evidence for an effect of the room condition on feeling awake.

Physiological stress

To gain insights in the estimated mean values RMSSD (based on three 5-minute samples, each 15 minutes in between) per each condition, these have been displayed in table 6. The highest mean RMSSD is reported in the hard/no smell condition, the lowest RMSSD was reported in the hard/smell condition. The lower the RMSSD the higher the indication of physiological stress (or fatigue).

Table 3: The mean and standard deviation of the physiological stress measurement RMSSD, per room condition.

		Hard	Soft	Total
No smell	Mean	53.33	50.48	51.68
	SD	16.24	20.73	18.85
Smell	Mean	47.04	49.52	48.26
	SD	17.23	18.41	17.67
Total	Mean	49.98	50.06	50.02
	SD	16.89	19.55	18.27

To test whether these conditions (smell and soft) have an effect on the physiological stress, a repeated two-way Anova was conducted. The residuals were not normally distributed (Swilk-test $p = 0.00$). Also transformations did not seem to improve the normality of the residuals, so a linear mixed-effects model was conducted. This is more flexible than a repeated Anova, and handles missing values better, which

this mean RMSSD data has. The model included ID as a random intercept to account for repeated measures. It was run with robust standard errors, adjusting for clustering within participants, as the normality assumption was not met either (Swilk-test $p = 0.04$). The overall model was not statistically significant, $\text{Chi}^2(3) = 3.80$, $p = .284$, suggesting that the predictors did not significantly explain variation in mean RMSSD.

The main effect of smell was not statistically significant, but only marginally ($\beta = -9.19$, $\text{SE} = 5.48$, $z = -1.68$, $p = 0.09$), suggesting that this data did not find evidence for trend where the presence of smell has a negative effect on lowering the RMSSD values (higher physiological stress). The main effect of softness was non-significant ($\beta = -8.92$, $\text{SE} = 6.50$, $z = -1.37$, $p = 0.17$). Similarly, the interaction between Smell and Softness was not significant ($\beta = 9.98$, $\text{SE} = 6.71$, $z = 1.49$, $p = 0.14$). This means that in this data, no evidence was found for an effect of the room conditions on the RMSSD.

A null multilevel model with random intercepts for participant ID showed that 70.3% of the variance in mean RMSSD was attributable to between-person differences ($\text{ICC} = 0.703$). A likelihood ratio test confirmed that a multilevel approach was appropriate ($\text{Chi}^2 = 86.79$, $p < .001$).

Time series trajectory appraisal

As it might be interesting what the trajectory of the physiological stress showed over the time spent in the room, the three time samples are visually appraised. Figure 10 shows the RMSSD trajectory over the three time samples, for each condition. All rooms show a slightly upward estimated marginal mean trajectory indicating participants showed lower physiological stress over time, on average. This could possibly suggest a restoration effect or habituation. Yet, it is also apparent that the height as well as the trajectory varies a lot per participant, which is also reflected in the high standard deviations in the table above.

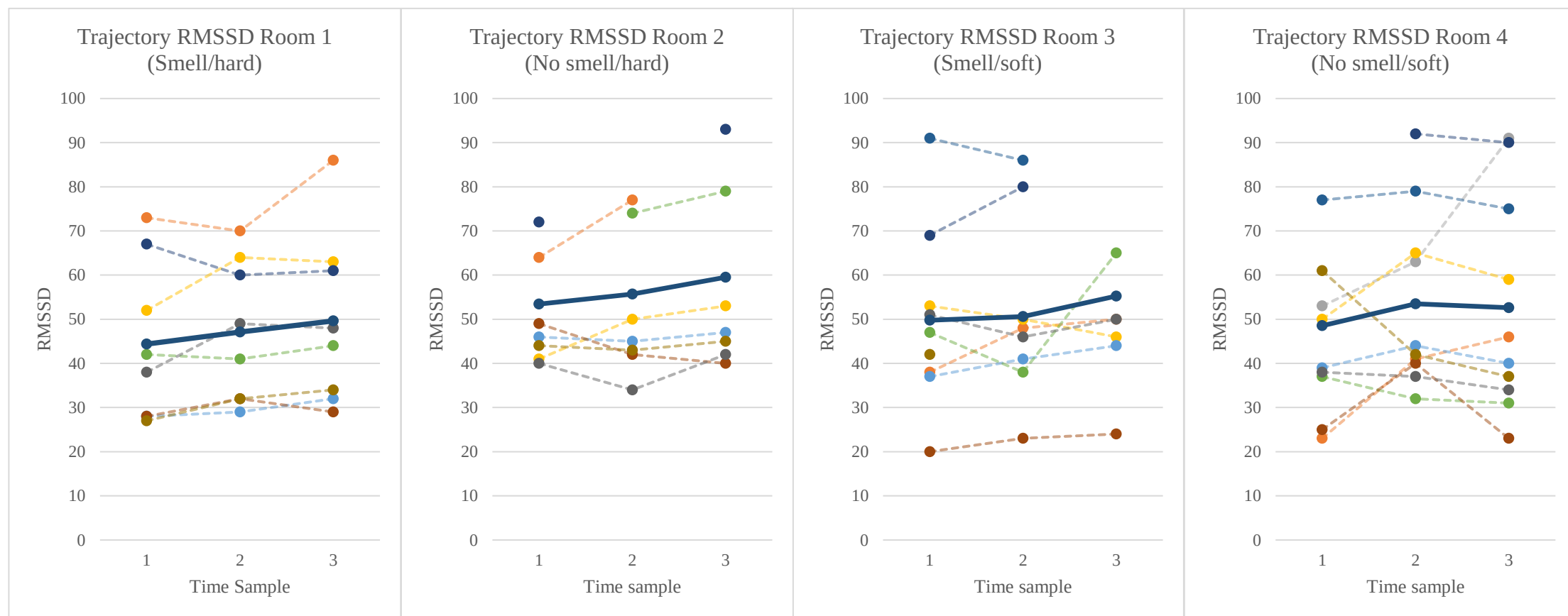


Figure 10: The RMSSD trajectory for each room. The dashed lines show the trajectory for each participant; the solid line shows the estimated marginal means of all participants. All participants are represented by the same colour for each of the graphs.

Correlations between psychological and physiological stress

To evaluate the correlation between psychological stress and physiological stress, three separate linear mixed models were run with the mean RMSSD as dependent variable and each psychological stress dimension as predictor, with ID as random intercept.

The model with tension as predictor met the normality (Swilk-test $p = 0.616$) and homoscedasticity assumptions. The model found a statistically significant effect between the mean RMSSD and self-reported tension ($\beta = 1.22$, $SE = 0.39$, $z = 3.14$, $p = 0.002$), indicating that as the reported tension increases, the mean RMSSD increases slightly too. This is the opposite of the expectations, as a lower RMSSD indicates a higher physiological stress.

The model with pleasure as predictor met the normality (Swilk-test $p = 0.540$) and homoscedasticity assumptions. The model found no statistically significant effect between the mean RMSSD and pleasure ($\beta = -0.74$, $SE = 0.64$, $z = -1.16$, $p = 0.245$), indicating that the data found no evidence for a correlation between the mean RMSSD and self-reported pleasure.

The model with awake as predictor has met the normality (Swilk-test $p=0.472$) and homoscedasticity assumptions. This model found no statistically significant effect between the mean RMSSD and the feeling awake ($\beta = -0.36$, $SE = 0.42$, $z = -0.84$, $p = 0.401$), indicating that the data found no evidence for a correlation between the mean RMSSD and self-reported feeling awake.

4.1.2 Room manipulations and Sensory appraisal analysis (quantitative diary), *Smell*

To gain insights in the estimated mean smell appraisal (7 point Likert scale, higher rating indicating more positive experience) per each condition, these have been displayed in figure 11 as well as the individual participant values. The highest mean smell appraisal is noted in hard/no smell condition (room 2). The lowest was noted in the soft/no smell condition (room 4). This figure shows that the least variance and highest values are given to the neutral (no smell) hard condition.

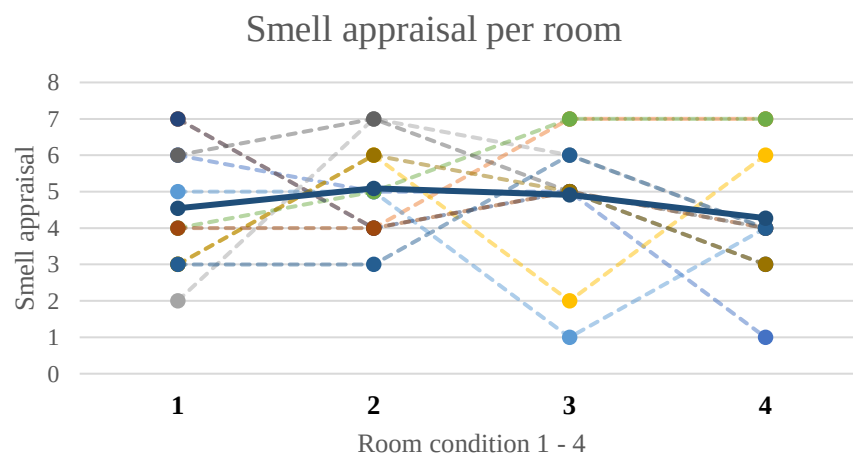


Figure 11: The mean and individual values of the smell appraisal. The dotted lines represent individual values, whereas the solid line shows the mean. The room conditions: 1(smell/hard), 2 (no smell/hard), 3 (smell/soft), 4 (no smell/soft).

A two-way repeated Anova was run to check for an effect of the room conditions on smell appraisal. The residuals followed a normal distribution (Swilk-test $p=0.41$). The model did not find any statistically significant results. It showed for the smell condition $F = 0.01$ ($p = 0.94$), for soft condition $F = 0.18$ ($p = 0.68$) and for the interaction between soft and smell, $F = 1.57$ ($p = 0.24$). This means that this data did not find any evidence for an effect of the room conditions on smell appraisal.

Touch

To investigate the estimated mean touch (and feel) appraisal (7 point Likert scale, higher value represents a more positive experience) per room condition, these, as well as the individual scores, have been displayed in figure 12. It depicts the highest mean in the soft/smell (room 3) condition and the lowest mean in the hard/ no smell (room 2) condition. There is again depicted some variance, especially in room 3.

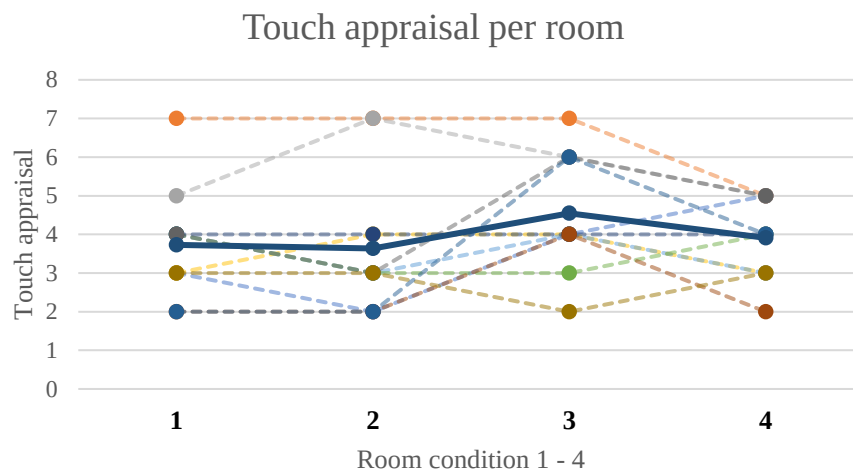


Figure 12: The mean and individual values of the touch appraisal. The dotted lines represent individual values, whereas the solid line shows the mean. The room conditions: 1(smell/hard), 2 (no smell/hard), 3 (smell/soft), 4 (no smell/soft).

To investigate whether the room conditions have an effect on the touch appraisal, a two-way repeated Anova was run. The normality of the residuals assumption was not met (Swilk-test $p = 0.002$), and did not improve with transformations. Therefore a linear mixed-effects model was conducted, as this is more flexible than a repeated Anova. The model included ID as a random intercept to account for repeated measures. It was run with robust standard errors, adjusting for clustering within participants, as the normality assumption was not met either (Swilk-test $p = 0.047$). The overall model was not found statistically significant and reported a $\text{Chi}^2(3) = 4.66$, $p = 0.20$.

The main effect of smell was non-significant ($\beta = 0.09$, $SE = 0.28$, $z = 0.32$, $p = 0.75$), suggesting there is no evidence for an effect of the smell on the touch appraisal. The main effect of softness was also non-significant ($\beta = 0.27$, $SE = 0.49$, $z = 0.56$, $p = 0.58$). Similarly, the interaction between Smell and Softness was not significant ($\beta = 0.55$, $SE = 0.53$, $z = 1.03$, $p = 0.30$). This means that in this data, no evidence was found for any effect of the room conditions on the touch appraisal.

Vision and auditory exploration

Although visual and auditory experience is not within the scope of the pilot study research question, it could provide interesting insights and context. Therefore, a quick overview of the means per room condition is added as an additional exploration of the sensory experience. These sensory modalities were not manipulated, and therefore, in a simple (non-multisensory) view, should not differ too much. Table 9 and 10 provides the estimated means of the visual and auditory appraisal per room condition. The highest mean visual appraisal (most positive) is noted in the soft/no smell condition, the lowest in the hard/smell condition. For the auditory experience, the highest mean is found in the soft/smell condition, and the lowest in the soft/no smell condition. Even though, these conditions were not manipulated, they do show different mean values per condition, showing different visual and auditory differences. But as they are not the main research objective, these have not been tested.

Table 4: The mean and standard deviation of the visual appraisal, per room condition.

		Hard	Soft	Total
No smell	Mean	4.73	5.55	5.14
	SD	1.31	1.70	1.56
Smell	Mean	4.36	5.18	4.77
	SD	1.69	1.78	1.77
Total	Mean	4.55	5.36	4.95
	SD	1.51	1.73	1.67

Table 5: The mean and standard deviation of the auditory appraisal, per room condition.

		Hard	Soft	Total
No smell	Mean	4.73	3.27	4.00
	SD	1.84	1.74	1.92
Smell	Mean	4.73	5.09	4.91
	SD	1.68	0.80	1.32
Total	Mean	4.73	4.18	4.45
	SD	1.75	1.63	1.70

4.1.3 Sensory appraisals and psycho-physiological stress analysis

Psychological stress

Tension

A linear mixed-effects model was conducted to examine the relationship between Tension as dependent variable and the predictors smell appraisal and touch appraisal, while accounting for repeated measurements using a random intercept for ID. It was run with robust standard errors, adjusting for clustering within participants, as the normality assumption was not met (Swilk-test $p = 0.007$). The overall model was found statistically significant with a $\text{Chi}^2 = 57.49$ ($p < 0.001$).

The analysis revealed a significant negative effect of smell appraisal on tension ($\beta = -2.29$, $SE = 0.85$, $z = -2.68$, $p = 0.01$), indicating that more positive ratings of smell were associated with lower tension. Similarly, touch appraisal also had a significant negative effect on tension ($\beta = -2.63$, $SE = 1.14$, $z = -2.30$, $p = 0.02$), suggesting that a more positive rating were linked to lower tension. These two effects are visualised in figure 13. The interaction effect between smell and touch appraisal showed a non-significant trend ($\beta = 0.35$, $SE = 0.18$, $z = 1.94$, $p = 0.053$), indicating that there is no strong evidence that the effect of smell appraisal on tension depended on touch appraisal (or vice versa). Overall, these results suggest that both smell and touch contribute to reduced Tension.

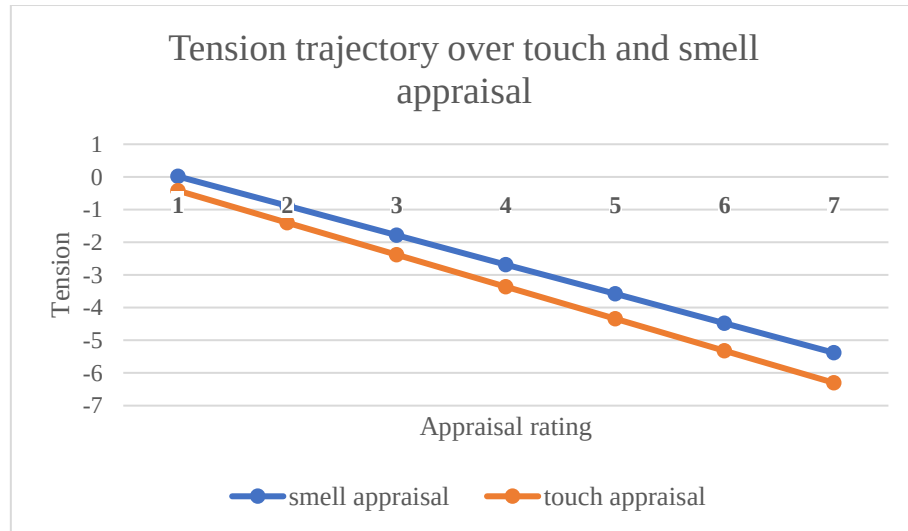


Figure 13: visualisation of the estimated margins of the effect smell and touch appraisal has on the tension dimension of psychological stress. Higher sensory ratings indicate a more positive experience.

Pleasure

A linear mixed-effects model was run for the dependent variable pleasure was well, also while accounting for repeated measurements by random intercept for ID. This model too was run robust, as it did not meet the assumptions of normality (Swilk-test $p = 0.01$). The overall model fit was found to be statistically significant with $\text{Chi}^2 = 38.93$ ($p = 0.00$).

The model showed a statistically significant effect for both the smell appraisal ($\beta = 1.79$, $SE = 0.34$, $z = 5.26$, $p = 0.00$), as for the touch appraisal ($\beta = 2.25$, $SE = 0.42$, $z = 5.40$, $p = 0.00$) on pleasure. Both positive coefficients indicate that a higher rating of smell or touch correlate with a higher rating of pleasure. The interaction between smell and touch appraisal too was found to be statistically significant ($\beta = -0.30$, $SE = 0.06$, $z = -4.68$, $p = 0.00$), indicating that the relationship between smell appraisal and pleasure depends on the value of touch appraisal (or vice versa). Specifically, the negative coefficient suggests that as touch appraisal increases, the effect of smell appraisal on pleasure becomes less positive (or more negative). This effect is showcased in figure 14. The first graph shows when touch is rated 1, a more positive appraisal of smell increases pleasure, but in the second and third graph, when touch appraisal increases, the starting point of pleasure (with low smell appraisal), is already high, and therefore flattens out.

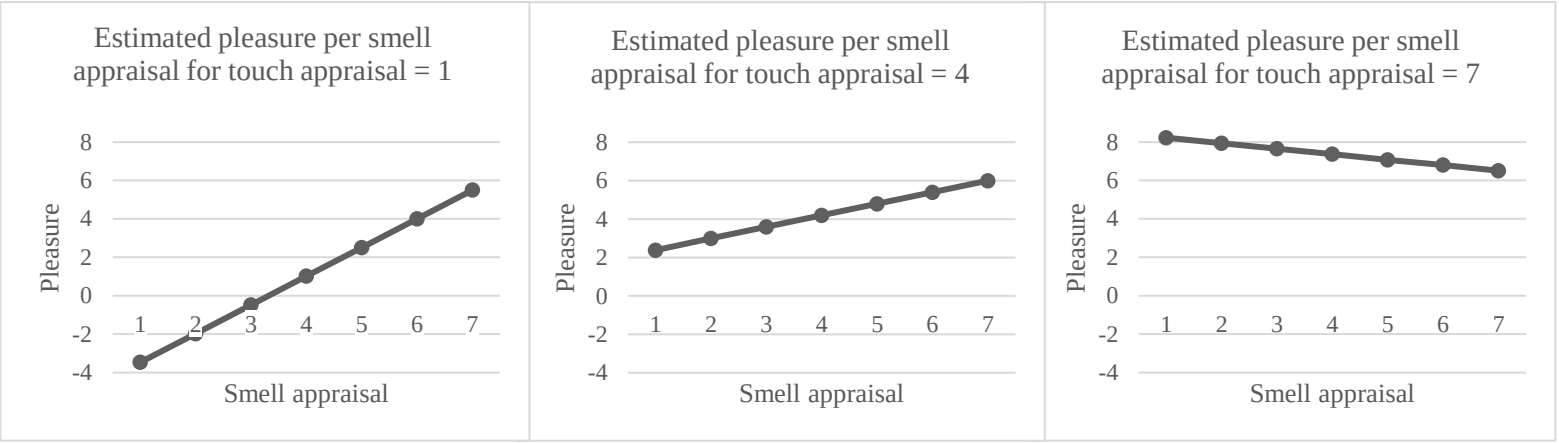


Figure 14: The estimated margins of the effect smell appraisal has on pleasure, per touch appraisal.

Awake

To investigate the effect of the smell and touch appraisal on the awake dimension, another Linear Mixed Model was run with ID as random intercept. The residuals followed a normal distribution (Swilk-test $p = 0.11$). The overall model was found not to be statistically significant with $\text{Chi}^2 = 5.87$, $p = 0.12$.

The analysis found no statistically significant effects for smell appraisal ($\beta = 1.91$, $\text{SE} = 1.05$, $z = 1.82$, $p = 0.07$), nor for touch appraisal ($\beta = 1.53$, $\text{SE} = 1.25$, $z = 1.22$, $p = 0.22$). Also the interaction effect was found not to be statistically significant ($\beta = -0.31$, $\text{SE} = 0.23$, $z = -1.32$, $p = 0.19$). This means that the data did not find any evidence for an effect of touch or smell appraisal and feeling awake, nor that the two sensory appraisals influenced each other's effect on the awake rating.

Physiological stress

To test for an effect of the touch and smell appraisal on the mean RMSSD (physiological stress), another linear mixed model was run, with ID as random intercept. This model too was run robust as it did not meet the normality assumption (Swilk-test $p = 0.01$). The overall model was reported to be not statistically significant with a $\text{Chi}^2 = 2.86$, $p = 0.41$.

The model found no statistically significant correlation for smell appraisal ($\beta = -1.07$, $\text{SE} = 2.32$, $z = -0.46$, $p = 0.64$), nor for the touch appraisal ($\beta = 3.04$, $\text{SE} = 4.03$, $z = 0.75$, $p = 0.45$). Also the interaction effect was found not to be significant ($\beta = -0.13$, $\text{SE} = 0.51$, $z = , p = 0.07$). This indicates that the model did not find any evidence for an effect of touch or smell appraisal on the mean RMSSD and thus physiological stress, nor that the two sensory appraisals influenced each other's effect on the awake rating.

4.1.4 mediation analysis

The previous analyses have shown that there is no statistically significant effect for path a nor for path c (see figure 6, chapter 3.6). Therefore, a mediation analysis is not necessary, as there is no indication of a mediation effect.

4.2 Qualitative analysis

The qualitative analysis of the interviews and focus groups provides insights into how participants perceived and experienced the experimental conditions. The applied coding framework and the frequency of occurrences can be found in Appendix H. The transcripts were kept in the same language as the interview or focus group was in (English and Dutch, depending on whether everyone in the group spoke Dutch). The snippets in the results have been translated to English. Where translations might not have precisely the same connotation, the original wording is presented as well.

Theme 1: Smell experience

1.1 Smell strength perception varies among participants

Participants perceived smells differently depending on their sensitivity and positioning within the space. The same scent was sometimes unnoticed by one participant but perceived as too strong by another. Smell was more likely to go unnoticed if it was introduced in the first room participants entered, and scents tended to fade from awareness over time. *'I remember thinking that it didn't completely smell like nothing like the other room. The other room which had no smell, I assume. So I remember smelling that it was a little bit different. But I couldn't like, honestly, if I didn't. If I wasn't in the room with absolutely no smell before, I probably would have said that it smelled like nothing, so.'*(p9)

1.2 The smell appraisal through personal association

The smell was often described as homely or like a 'department/home-goods or fragrance' store. These personal associations could influence the experience of the scent, *'When I'm at my parents' home, there is always a scented candle lit. Or the fireplace is lit in the winter. So I am really used to strong scents being present. So I thought it was rather nice. I only thought, I'm studying at my parents place'* (p3) as well as the experience of the room *'I really think that a room is more cosy (Dutch: gezellig) with the scent present'*(p7). Although this relaxed association was not loved by all participants in the particular setting *'I think it is also a bit distracting. Or maybe... It can be positively distracting, like it makes you more relaxed. But that is not really appropriate'* (p13). These associations also had those effects in a more negative point of view. *'It smelled a little musty. First of all, it is not nice to enter a musty room. Secondly, it also makes you drowsy and tired (Dutch: Muffen en inkakken) as well'*(p1).

1.3 The smell appraisal through smell strength

When participants rated a scent positively, they described it as just "nice," whereas negative ratings were often strongly unfavourable. *'When you don't smell anything, that is super positive. Because you don't experience any negative.. I'm thinking, when it comes to smell, when you don't notice it, it is good. Because there is nothing stinky or distracting, especially when studying.'*(p7). It was quite apparent that the absence of the smell was also rated as positive, as well as a slight scent that would tune out over time. *'I don't mind the smell. It was just like something I noticed but it kind of went away in a bit. I kind*

of got used to it quickly, so.’ (p15). Yet, when the scent was experienced as very strong, people had more issues with tuning it out, *‘It was a nice smell, but it was very present. It was so strong I could not get used to it’* (p7), and often experienced it as more negative *‘I really couldn’t handle the scent. It really was not for me.’* (p4) To even physical discomforts *‘it was so strong that I can feel it itching in my nose.’* (p9).

Theme 2: Touch experience

2.1 Physical touch, the joys of softness and annoyance of slippery items

The physical touch aspects participants reported on, were mainly the items around the desk. Soft textures were generally experience positively, especially the blanket on the chair. *‘The blanket on the chair, in the first room I didn’t use it. I was already very warm, so I didn’t need it. But I liked having something on my lap. It was like a soothing behaviour for me* (p3). But also without intent interaction, they liked having a soft blanket in their back *‘When it hit my back, I thought ooohh nice soft blanket’* (p6). Also the softness of the table cloth was sometimes rated positive *‘the cloth on the table was nice. In a very simple way, because it is softer... when you place your arm on the edge of the desk the whole time, because of the laptop, it is not on a hard edge the whole time.* (p11). Yet, some also experienced the tablecloth as annoying, mostly due to the slippery movement *‘I thought the tablecloth was very annoying. Because when your laptop was on it, and it moved around.. It was also placed over cables which made that it wasn’t laying down straight. Something should not distract, something that moves when I move. For example a mousepad would have been fine, that doesn’t move when I move my mouse.* (p6). Also the size of the desk varied from the soft rooms (smaller) than in the hard rooms, which was noted. Smaller desks were experienced as more annoying *‘That desk was very small. I’m a person that uses both laptop and an iPad. And then I also have my water and a keyboard with me.* (p3)

2.2 Embodied touch appraisal and other visual aspects

Many of the visual aspects in the room were placed outside of the participants reach, but were still experienced through embodied touch. *‘So in a way, there was less that I physically touched, but more the ‘Oh I could nicely lay in the couch and that would feel nice’*(p1). Especially the couch was often mentioned and talked about looking comfortable. This embodied experience was also sometimes confirmed *‘it looked like it could be comfortable. When I tried it out, it was comfortable.* (p15 about the couch). This embodied experience already started when entering the room, giving a very quick sense of the room *‘I knew I walked inside and saw, I thought, O how nice, a blanket!* (p5). Another visually haptic aspect mentioned was the feel and structure of the space, especially concerning the table cloth. Whether this was positive *‘It was kind of nice to for example place some other stuff on it (the table cloth) or make a bit of a distinction in your workplace’* (p11) Or negative *‘Maybe it also takes up space*

visually, from your whole desk.' (p5). The visual embodiment of the touch experience mostly led to atmospheric experiences through personal associations.

3.3 Touch appraisal and personal associations and emotions

Softness was commonly associated with a homely atmosphere, often through simulated touch. It frequently gave a positive feeling and experience, *'I thought the warm rooms also felt warmer. You are more at ease there'*(p7 about soft environment) Or it even made them feel more safe *'It feels safe when there is a couch with stuffed animals on it'* (p3).. This could also evoke feelings of relaxation or even the urge to lie down or take a break or even a nap, which was sometimes reported on as positive *'It makes it very homely. I get more motivated by it, because then I see a couch and I think 'Oh, I could lie on that when I'm done''* (p10), but also sometimes as inappropriate for a work/study setting or demotivating *'This (relaxing behaviour) is associated with the environment (soft room), so you are more easily inclined to relax and maybe work a little less hard* (p11)'. Also, the direct feeling of softness, for example of the blanket, was associated with homely and relaxing surroundings *'At home I also often take an extra pillow on my lap. I don't know, it just feels nice. So, when I'm at home it is the first thing I do, I think it is just relaxing.* (p3). But in that focus group conversation others commented, this soothing homely association had also been experienced as inappropriate for the setting *'I think it is lovely too, but not when I'm studying. The I know I just don't want to study anymore, get cosy underneath a blanket* (p7). The hard environment on the other hand, although less talked about in terms of touch, had more associations with office like environments. This touch experience was not always met as positive *'It felt very static and cold'* (p7). A few participants, mostly female, mentioned feeling cold, which they found distracting.

Theme 3 : Stress

3.1 Calmness induced by the rooms

Stress was not often explicitly mentioned by participants. Instead, feelings of calmness and relaxation were noted more frequently, which is often regarded as the opposite. *'For me it (soft furnishing) was the first thing I noticed. And that was the first thing that brought me rest. I immediately had a feeling of peace (Dutch: rust)'* (p3 on soft room). Which was earlier also mentioned in the soft homely warm associations of the soft rooms, as well as with the soothing behaviour of the blanket on the lap. *'Concerning productivity I didn't feel the rooms made much of a difference. But I did notice it being more chill in the soft rooms. Again with the idea that it is more atmospheric'* (p10). Even though the participants did mention relaxing feelings towards some of the room design elements, they often were sceptical about whether the environment could induce stress. Participants used more positive wording, but when asked directly about stress, most responses suggested annoyance or lack of focus rather than acute stress. The word stress seemed too strong to use. *'Maybe because I didn't have any real stressful*

activity today. Productivity maybe a little bit, but that was more... for me it was more clear that that was because it was the end of the day (p1 when asked about his stress).

3.2 Stressing factors

Although, the word stress was not often attributed by the participants, the cases when stronger negative emotions were communicated, it was mostly due to the smell being too strong, but strong wording was hardly ever mentioned. *'Room 1 is the bottom one for me, because I really couldn't handle the scent'*(p4). Also, for the table cloth, some annoyance was mentioned *'And I thought the tablecloth was terribly annoying, laying on the table'*(p6). Yet, when asked whether this was stress inducing, or they felt more stress in this room, they mentioned not to feel that much stress or were sceptical whether the stress was induced by the room, but attribute it more to other factors such as their activity *'Researcher: I'm interested in your stress levels, can you elaborate on how they were and how they might have differed per room? P4: My stress is generally low as I only have one course at the moment... Only in the 4th room I was reading a super exciting suspenseful chapter of a book which might have made me stressed'* or attributed stress to frustration towards themselves due to a lack of focus *'In room 4 (soft, no smell) I thought, I'm stuck. So I noticed to get more stressed, because I thought I really have to get on with it'* (p10). The personal association of the room also was mentioned to induce irritation indirectly *'The I think, no, well shit, why can't I do it there (on focussing in the hard room)? Then I get a little irritated more quickly. But I think that I conditioned myself that I have to perform well in bare (Dutch: kale) rooms.'* (p3)

4: Other Findings

4.1 Effort put in the room is positive

Some participants elaborated that effort being put in the environment has for them a positive effect, mediating a more positive experience of an environment. *If I were to relate this to somewhere outside, I would also feel like, yeah, I do appreciate it when people at least make an effort to make the space nice. If you have one of those awful office spaces, you might as well do something with it.* (p1) This can be in multiple forms. In this pilot has both been said about the scent as the decorations in the soft rooms *'Precisely when there is a scent, you know that some attention has been given to it, and that the room will make you feel a bit more comfortable because you know that some effort has been put into it'* (p.11).

4.2 Individual differences in positive stimulation or distraction

There seemed to be individual differences in preference for distractions were evident: while some required a plain, minimalistic environment to focus, or even the personal association of an office, others needed slight stimuli to avoid becoming distracted by larger disruptions. For example, *'I found all the decorations in the rooms to be really distracting while studying. And in the basically empty rooms, I*

was. *I could really focus really well*' (p9). Meanwhile, the other participant in this focus group argued *'Because just studying in an empty room with absolutely nothing interesting to see while doing, like, very, you know, not particularly, like, you know, vibrant work, just, like, reading and taking notes and all that. And also just in dead silence, my mind was screaming for anything. Like, any kind of stimulus. So when I was actually, more comfortable in rooms that I had some stuff to look at or actually nice smells, it did feel a bit better.'* (p15). This need for stimulation was even more sought after in the context of working there for longer period than an hour *"So for your psyche, I would say it's better than spending the whole day in a bare room where you're just staring at a white wall or a white pillar"* (p.11). Homely associations elicited mixed responses—some found them calming, while others felt they hindered productivity by making them too relaxed, as also discussed in the previous sections.

4.3 seating arrangements induced social cues

Seating arrangements also influenced perceptions; soft and well-used chairs were often seen as social spaces, which was generally viewed positively. *'In the other two rooms (soft rooms), it actually felt like, you know, like someone actually, you know, worked there or studied there. Someone actually took the effort'* (p15). This social cue also seemed to be influenced by the arrangement of the seatings *'I found it really strange if I didn't take care of this, but basically, the chairs were positioned in such a way that it really looked like two people had just gotten up. (p4) R: okay, and was that a positive or negative thing? P4: Yeah, kind of like that cozy, homely feeling that you recognize, you know? It just gives a good feeling. Yeah, okay.'*

4.4 A need for balance and variety in environments

Several participants emphasized the need for balance or a change in scenery depending on their work tasks. *"I think, on the one hand, that I do like having an environment where you can work well. But I think it's more about having a bit of variety that's good. And also, that you associate that variety with the space you're in. So, for example, if I say, okay, I've worked for an hour, I want to take a 15-minute break. Then, ideally, I would leave a somewhat plainer room to go to a warmer environment for a while. And vice versa, if I've just had my break in a cozy space, I would then go back to a slightly plainer room to continue working."* (p11)

While proprioception and ergonomics were occasionally mentioned in conversation as a remark on the room, they fell outside the scope of this pilot study.

4.3 Results summary

Quantitative results

None of the room conditions (smell or soft materiality) showed to have any statistically significant effect on either one of the psychological stress dimensions, nor the physiological stress mean RMSSD. The physiological stress (RMSSD) unexpectedly correlated negatively with the tension dimension,

indicating that higher tension would be correlated with lower physiological stress. The room manipulations, smell and soft, also showed not to be statistically significantly correlated with the sensory appraisals of smell or touch (/feel). The trajectory seems to show a declining physiological stress over the time spent in the room, for 3 of the 4 rooms. Yet, this is not statistically analysed.

The only statistically significant findings this quantitative analysis did find was on the effect of the sensory appraisals of both smell and touch on the psychological stress dimensions tension and pleasure, indicating more positive appraisals showing lower tension and higher pleasure. The interaction of smell and touch appraisal also was statistically significant for pleasure, indicating that the relationship between smell appraisal and pleasure depends on the value of touch appraisal (or vice versa). Specifically, the negative coefficient suggests that as touch appraisal increases, the effect of smell appraisal on pleasure becomes less positive (or more negative). The dimension awake or physiological stress (RMSSD) did not show significant effects.

As two of the three paths, as per the mediation diagram, were not statistically significant, there is no indication of any mediation. Therefore, a mediation analysis was not performed.

Qualitative analysis

Smell played a significant role in how participants perceived their environment, through perception, personal association and strength. The presence of a distinct scent was sometimes associated with care and intentionality or comforting personal associations. Yet it could also have very negative effects when the scent was too strong and/or disliked. The negative effects seemed more impactful than the positive. Also, the absence of any (bad) smell was considered positive.

Touch and texture were also important factors in shaping participants' experiences. Soft and comfortable elements were often preferred over hard, plane surfaces, as they contributed to a sense of relaxation and comfort. Yet this was not always appreciated. Some felt that that association was not appropriate or were a bit annoyed by distractions or moving elements.

A key theme was the psychological effects of different environments, particularly in relation to stress. Stress was not often mentioned; participants talked more about calmness (the positive opposite) or used less strong words such as annoyance. The calmness was mostly related to the soft and warm environments. Possible stressors were strong smells, their inability to focus (while being in a room/setting where one is ought to focus), or their own tasks. This was not often contributed to the room. Whereas the positive calmness was noted to be linked with the environment.

There were some additional findings such as the positive effect of the idea that someone put thought in an environment, the need for variation in the work environment throughout a day, the positive notes of traces of human presence, and the big individual differences in over- or under stimulation of an environment.

Overall, the analysis suggests that participants have very different opinions and needs for a working environment for various reasons. This could conclude in their appreciation for an environment that balances functionality with warmth or a variation of space throughout their day, depending on their activity.

5 DISCUSSION

This thesis work made use of a wide arrangement of methodologies, to learn how sensory ethnographic methodologies could add additional insights on workplace research, more specifically concerning psycho-physiological stress in relation to touch and smell. The strength of this study is that it used both quantitative techniques to measure effects, and qualitative ethnographic techniques to find explanations and context of the effects found in the quantitative data. This study was performed with an explorative nature, as these methodologies are not often combined, and these sensory modalities are not often researched in the physical workplace environment. This exploration has provided some interesting insights in both the effects and mechanisms of the sensory influence on psycho-physiological stress, and in employing sensory ethnographic and physiological techniques in workplace research. Key findings include: (1) the liking/appraisal of smell and touch is highly subjective, but plays a big role in psycho-physiological stress, (2) homely associations may lead to (too much) restoration, (3) The touch experience does not have to be physical but can be visually simulated, (4) Accurately conducting and interpreting physiological stress measurements presents considerable methodological challenges, and (5) there seems to be a dissonance in performing ethnography in a lab setting. After, these discussion points have been explained, the main learning goal of (6) what can we learn about applying ethnographic methods in workplace design and psycho-physiological stress can be discussed.

5.1 The role of touch and smell appraisal

The findings of this study underscore the importance of sensory appraisal—the way individuals interpret sensory stimuli—in shaping psychological stress responses. While the room manipulations (smell and soft materiality) showed no direct effect on psycho-physiological stress, participants' subjective experiences of these stimuli did relate to psychological stress dimensions, particularly tension and pleasure. These results showed more positive appraisals correlating with lower tension and higher pleasure. This highlights that it is not just the presence of touch and smell alone, but how they are perceived and appraised may also determine their impact.

As supported by previous literature (Bentley et al., 2022; Matsubara et al., 2023; Spence, 2020b), sensory experiences are highly personal and context-dependent. This was echoed in the qualitative data: participants rated the same sensory environment in vastly different ways, often grounded in personal associations. Spence (2020b) explains this as a very common mechanisms, where people place positive or negative associations on smells through associative learning through earlier experiences. Although the citrus scent was chosen based on literature suggesting its general pleasantness (Liu et al., 2023; Spence, 2020b), participants' responses ranged from feeling comforted to distracted or demotivated—with ranging associations such as 'homely settings by the fireplace', 'baking cookies' or 'department stores', all experiences participant are likely to have learned to associate the smells with through memory or culturalization (Spence, 2020b). Similarly, the soft materiality elicited both relaxation and

discomfort, depending on whether participants associated it with a cozy home or an unfitting work atmosphere. These findings align with research showing that touch perception is also influenced by prior experience and context (Crippa et al., 2012; Plaisier et al., 2023), though literature on the link between touch and stress remains limited. As these experiences can be very much varying between different people, workplace designers might want to refrain from using very strong sensory features such as a strong smell. Sticking to more neutral, and generally positively associated cues could be a more appropriate course.

The empirical study findings suggesting that positively rated sensory inputs can reduce psychological stress, are corroborated by literature (Liu et al., 2023; Matsubara et al., 2023; Spence, 2020b). However, no associations were found between sensory input and physiological stress (RMSSD), which was unexpected given that other research did find evidence linking pleasant scents with HRV (e.g. Liu et al., 2023). A possible explanation, besides the small sample size, is that this study only used RMSSD as a physiological marker, which may also reflect cognitive load, fatigue or arousal (Srinivasan et al., 2024). Higher levels of fatigue as well as arousal are typically linked to decreased RMSSD values, indicating reduced parasympathetic activity and a heightened physiological stress response (Huo et al., 2023; Srinivasan et al., 2024). Moreover, it is good to note that physiological stress often finds low consistency over different studies (Schuller, in press) and psychological and physiological stress often show weak correlations (Ginty et al., 2017).

5.2 Homely associations and (too much) restoration

The study found that the soft environments, and the smell created homely associations for many of the participants. As Chatterjee (2021) explained, the participants described in the focus groups how the smells, soft textures or furnishing reminded them of similar experiences in home environments, creating homely associations. They reported to feel more at rest and calm in those environments, especially concerning the soft materiality. Other studies, too, seemed to find similar effects. Graham et al. (2015), suggest that environments emphasizing comfort, togetherness, and intimacy can significantly reduce stress and improve mood by providing a sense of security and emotional support. But what their study also found is that this need varied per room that they studied. For instance, home offices and study areas, designed for productivity and stimulation, were found to enhance focus and concentration. The ambiance in these rooms is tailored to support cognitive tasks and minimize distractions, thereby improving concentration and academic performance (Graham, Gosling, & Travis, 2015). This also reads into the findings from the focus groups. Many participants reported feeling calmer and at rest in the rooms described as ‘homely’, yet what some of them also reported is how they did not like that for the specific context of a working/study area. They felt less focussed, less productive which could in some cases then again increase stress or annoyance. This effect can also be explained by the Yerkes-Dodson law (1908), which posits that there is an optimal level of arousal (stress and alertness) for peak performance. According to this law, moderate levels of stress can enhance focus and productivity, while

too little or too much stress can impair performance. This inverted-U relationship suggests that a certain amount of stress is necessary to maintain alertness and motivation, but excessive stress can lead to burnout and decreased productivity (Cohen, 2011; Yerkes & Dodson, 1908). When the stress becomes too excessive, it becomes problematic, but too much restoration is not fitting for a workplace context. A workplace should try to facilitate that optimum level of arousal for employees to be able to focus and be productive. The focus groups of the empirical study also mentioned that the feeling of focus and productivity in that setting, is positive for their overall mood and subjective wellbeing. The workplace environment should therefore not be a stressor, but should also not have too much stress restoration effect, for example via homely associations. What those exact design aspects are, are again based on personal associations and should be researched more.

5.3 The role of embodied touch

In the empirical study, the soft environments were also decorated with soft textures that the participants did not physically touch. Still they did mention for instance how the couch looked comfortable or how they looked forward to touching the blanket in the chair as it looked soft. Spence (2020a) also mentions that the tactile element is important, whether an object is touched or not. He states that material choices can evoke a sense of tactility, as the viewer or occupant mentally simulates the sensation of touching or caressing a visually engaging surface (Gallese, 2017; Spence, 2020a). It is interesting to see how this simulated touch, could still evoke emotions and could maybe also be stress reducing. This study did not particularly set out to measure these simulated effects, but they did emerge in the focus group conversations and seemed to have played a role in the experience of materiality. Ebisch et al. (2008) also showed neurological similarities between touch observations and being touched themselves, especially in intended touch. It might also be interesting to look into the relation with the 'look and feel' aspects of office design, which is researched more often (Al Horr et al., 2016; Kropman et al., 2023). When look and feel are described they are mostly described as visual aspects such as aesthetics, but the embodied touch might play a role in this. This was also proposed by Gallese (2017, 2019), where he argues that embodied simulation offers a fresh perspective on aesthetic experience and can be considered an interesting link in bridging neuroscience and the humanities within the biocultural framework. These ideas now mainly stem from artistic practices and ideas (Gallese, 2017; Hayes & Rajko, 2017) but could be interesting to investigate how they might translate into workplace design. These ideas call for more cross modal and multisensory research and design approaches (Spence, 2020a), also investigating the interaction of the senses and how they might play a role in aesthetics and ambiance.

5.4 Challenges in Measuring Physiological Stress

This empirical study found some unusual findings concerning the physiological stress data. Not only did it not find any statistically significant effects with any of the predictors, even where psychological stress did, it even had a negative relation with psychological stress. This would indicate that the higher

the psychological stress, the lower the physiological stress. It is not too strange that psychological stress and physiological stress show different results, although interlinked they are distinct concepts (Ginty et al., 2017), but the negative relation with psychological stress seems irregular. This thesis showed that there is a need for expertise to perform physiological measurements well and reliably. A lack in expertise and confounding factors could lead to less reliable and accurate results (Schuller, in press). The study found that the analysis process itself too is very time consuming and requires expertise to do well. Furthermore a highly controlled study was necessary for reliable data, due to many probable confounding factors. These can be both difficult to acquire in 'real setting' workplace research, but necessary for reliable results. Many other papers too seem to struggle to find good quality, consistent and replicable physiological stress measurements in workplace design (Schuller, in press). She, for instance, argues that there is a growing interest in wearable physiology measurement equipment, but workplace researchers should be mindful of the complex biological nature of physiological stress and the limitations of applying physiological measures in real-world settings. Although tools like HRV and EDA could be well-suited for field research, their reliability can be compromised by confounding factors present outside controlled environments (Schuller, in press). Although HRV and EDA are one of the most used measurement techniques for physiological stress (Valentine, 2023), it might also be of interest to investigate other measurement types such as blood pressure, respiratory rate or body temperature (Valentine, 2023). Yet, it could very well be the case that these methodologies would propose similar difficulties because of the highly complex biology (Schuller, in press). The ethnographic methods could possibly help explain certain context for possible effects or lack of effects, this is probably more the case for psychological stress. As discussed in the literature review, physiological stress can is not always experienced consciously (Becker et al., 2023). This is why physiological measurements done reliably and well, is still very important.

5.5 The dissonance of ethnography in a lab setting and how to balance

Ethnography, as described in the literature section, is where the researcher also immerses themselves in the experience and tries to research the real and direct environment and sensory experiences (Pink, 2015). Performing ethnography in a lab setting is therefore inherently contradictory. This lab setting was purposely chosen to still be able to manipulate the environment, to investigate certain sensory modalities, without other confounding factors and to help the physiology measurements. Other controls such as prohibiting coffee intake and semi controlling for tasks, were necessary for gaining more reliable physiological data in this lab setting. This too is in contrast with the ethnographic ideology and might infringe on the ecological validity. But it reflects on the difficult and nuanced balance of trying to marry the two methodologies. In future research this balance should be considered well and formed accordingly to the main research objective.

Also, this study did not take the researcher as first person perspective, but merely analysed the participants responses alone. Although, this, as well as the combination of more quantitative

measurements, has been done more often under the title of ethnographic methods (Bolger et al., 2003; Clarkson & Hodgkinson, 1999; Mazzola et al., 2011; Warren, 2002), whether it really is ethnography in the first essential definition might be debatable. It did work out well to take the participants perspective as main perspective, as the data showed a wide variety of experiences and personal associations and opinions. This might not have been as clearly shown if the researchers first perspective would have been the main perspective and could therefore be a good approach in highly subjective experiences. The researcher was still involved in the environment and experience, in the sense that the researcher designed the environment to their (and the supervisors) interpretation of what soft and hard environments would constitute of, and what scent would be nice and appropriate. This resulted in a subjectively designed environment, engaging the researcher in an ethnographic sense. This was also done in earlier work-place ethnographic studies (Warren, 2002, 2008). The environment design was based on the researchers own experiences, which was then in the empirical study reflected on by the participants. Although most of them let know that they knew that the researcher designed it, they did not seem to have difficulty expressing critiques on the environment. This could be due to the student knowledge that it was university equipment, or that experimental environment manipulations often do not per say reflect the opinions and likings of the researcher, but are based on the study's research aim. Another way the researcher was involved in the ethnographic process was via the subjectivity in coding. Although, it was tried to perform the coding and thematic analysis as objective as possible, it is very difficult to refrain from any subjectivity in that process.

5.6 What can sensory-ethnographic methods add to workplace design research?

From the literature review was learned that ethnographic methods has been applied in workplace design research (Warren, 2002) and workplace stress (McGibbon et al., 2010; Soucek et al., 2023), but not often in sensory sense and not often combined with psycho-physiological stress. Yet, there did seem to be indications that it could have potentially interesting insights to combine these quantitative and qualitative measurements (Mazzola et al., 2011; Soucek et al., 2023).

Previously, stress was often not a main research goal in sensory ethnographic research, especially concerning workplace design, but it did sometimes came up as a topic (Clarkson & Hodgkinson, 1999; Glaw et al., 2017; Golvani et al., 2021). This thesis learned that you could use it as main topic and research it more focussed, but does call for keeping the conversations open to context and nuance, as that is the main strength of sensory ethnography (Pink, 2015). As established before, the combination with physiological stress might be more difficult to perform in workplace research. Although it would have been interesting to be able to get the quantitative and qualitative data really side by side, this might prove to be difficult. This thesis did find a novel way how the data could enrich each other none the less. The ethnographic methods were able to provide rich context, reflexivity and highlighted mechanisms of psycho-physiological stress effects based on. This was already speculated based on the

literature review where it was applied in different context and research goals (Pink, 2013), and confirmed in this domain as well.

Diaries, while already known for capturing in situ information (Bolger et al., 2003; Reid et al., 2011), also proved to be valuable tools as probes in the mediated focus groups, prompting reflection and memory during interviews. They could enhance the depth of insights and help the researcher structure focus groups better, with more targeted questions. This method could therefore possibly be considered both effective and easy to use for workplace research, especially when combined with interviews and focus groups. Regarding diary questions, literature shows a mix of open and more directed formats (Bolger et al., 2003; Mazzola et al., 2011; Reid et al., 2011). For lesser-explored senses, more guided questions might be more effective, as the more open questions often led to vague or visually-biased responses. In both real-world and lab workplace settings, participants could benefit from some guidance to access and articulate subtle sensory experiences that may not be top-of-mind (Pink, 2015).

In the literature review, it was established that an important part of sensory ethnography is trying to gather the information in situ, while experiencing the sensory encounters (Pink, 2015). Yet, this might not always be possible in certain settings or experiment designs. This study has showcased, that although it is important to direct the participants to the sensory modalities and experiences, recalling it aided by visual prompts and embodied experiences (Gallese, 2017) can still provide very valuable insights and still help participants recall and explain their experience. This is where the ethnographic method could really add to the more traditional methods, by guiding the participants through their sensory experiences. This seems especially key when researching lesser consciously experienced senses such as touch and smell. Although this method did not guide the participants as much as for example in smellwalks (Low, 2015; Pink, 2015; Sparkes, 2009), but it was already noticed that the open-ended, not sensory steered questions, were very visually or ambiance focussed. The more sensory focussed questions already seemed to help guiding the participants, but the most insightful information was gathered in the focus groups, where the most guidance, directed questions and follow-up questions would be asked.

All in all, the ethnographic methods of diaries and focus groups/interviews did seem very applicable and appropriate, for lab settings but also for in situ workplace studies. How the balance should be in lab setting vs in situ setting depends on the specific research goal. The lab setting can uncover more specific effects and can be more easily controlled which is especially practical with physiological measurements, whereas in situ research can give higher ecological validity.

6 Conclusion

This study explored how sensory ethnography can enhance research on workplace design and psycho-physiological stress by focusing on the underexplored senses of smell and touch. The literature study found that there is already quite a body of work concerning visual and acoustic influences, but smell and touch, although possibly showing promise of influence on the psycho-physiological stress levels of employees, has not been researched as much. The empirical study highlighted that smell and touch design might influence psychological stress, but primarily through personal sensory appraisal of the sensory experience rather than direct sensory input—highlighting the importance of subjective interpretation. The literature and the study demonstrated that sensory ethnographic techniques, such as sensory diaries and focus groups, can effectively capture nuanced, in-the-moment experiences of smell and touch. However, these have rarely been integrated with physiological measures like HRV, which proved difficult to perform as well as interpret reliably in this thesis. The study elaborated on how sensory ethnography adds rich, contextual insights beyond what quantitative data alone can provide. This holds especially true for senses that are less actively consciously experienced, such as touch and smell. It revealed how sensory stimuli are interpreted through personal associations, how “homely” design elements can both calm and distract, and how even visually simulated touch may shape experiences. Together, these findings support the value of incorporating sensory ethnographic methodologies in workplace research, especially when studying subtle and subjective phenomena like sensory modalities and stress.

6.1 Limitations

This study provided valuable insights into the impact of sensory environments on psycho-physiological stress, yet several limitations should be noted. The biggest limitation was the small sample size, which resulted in potentially lower reliability and smaller chances of finding statistically significant effects. Secondly, the environmental conditions were not fully controlled: visual stimulation differed between rooms, one room was noted to be affected by external noise, and the “no-smell” condition may have been compromised by a reported musty odour. These inconsistencies may have influenced participants’ experiences and affected the clarity of the manipulations. The environmental set-up was also based on rather limited existing insights and mostly the opinion of the researcher (and supervisors). This could have led to less optimal staging, and a bigger variance in appraisal of the conditions. In terms of physiological measurements, as mentioned in the discussion, HRV analysis required significant expertise, particularly in preprocessing and interpretation, and the use of only RMSSD limited the nuance of stress assessment. EDA data was excluded due to lack of time and expertise, as well as probable data noise caused by hand movements. Strict controls over participant behaviour, such as restrictions on caffeine or music, were necessary for physiological reliability but reduced ecological validity. Similarly, conducting all conditions in one day introduced potential fatigue or order effects, which randomization could not fully eliminate—especially given the small sample size. But this was

also necessary to control for daily variances in psycho-physiological stress. This study also did not take sensory interactions into account in the measurements, even though the results and other work suggests that this might play a role in the environment experience.

6.2 Future research recommendations

There is still a lot unknown in how different senses influence the psycho-physiological stress in the physical workplace. Future studies that mean to combine ethnographic techniques and quantitative measurements like this study, should first clearly define their main research objective and adjust the balance between controlled lab setting and ethnography with ecological validity accordingly. There should be a better investigation how physiological measurements can be applied more reliably in the workplace setting. Furthermore, pre-testing sensory elements—such as scent intensity and pleasantness—could ensure more consistent and positive experiences across participants, while involving users or design experts in defining environmental qualities like “softness” can reduce subjective bias. From the discussion we learned that homely associations have effect on calmness or could even lead to too much calmness. Interesting research would be what design attributes make for a homely environment, and what the optimal level of ‘homeliness’ and restoration would be for an office environment. This could also be studied in how it differs per room function, such as an office, meeting room or break room. Also, multisensory and cross-modal research could help gaining a better understanding in how we experience the workplace environment, for example with simulated touch or sensory congruency.

6.3 Implications for practice

This study illustrates the importance of considering a broader range of sensory factors—beyond the visual and acoustic—when designing workplace environments. Elements like material softness, ambient scent, and embodied, haptic experiences can all shape how people feel and function in a space. However, workplace design should strike a careful balance: while spaces that can help restore and reduce too much job stress, spaces that are too stress-reducing due to too much homeliness and/or softness, might not be appropriate either. This could potentially diminish focus, alertness, or productivity. It could be a good idea to make differentiation in different rooms with different functions. This would also help in the fact that the effects the environment can have are highly individual and subjective. If possible, giving employees an option in what kind of environment to work in, provides them a sense of control and can help them find their own optimal level of restoration and stimulation. Concerning smell, it was found that the absence of smell was already rated as highly positive. Artificial smells, even ‘nice smells’, can be experienced very differently. When experienced as negative, the effect on their mood seems to be highly negative. Concerning touch, design decisions should move beyond surface-level aesthetics or technical functionality and start to consider materiality in a more holistic way—how materials feel, how they are experienced physically or embodied, and how they influence perception and emotion. There is still much to uncover about the complex interactions

between sensory design, cognition, and stress, particularly regarding less-studied senses like smell and touch. These findings advocate for a more multisensory and human-centred approach to workplace design that recognizes the nuanced and individual ways people experience their environments.

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Appendix A – Procedure participant time schedule

Time	Activity
09:00	Welcome in room 1
09:15	Participant working
10:15	Participant diary and EMA
10:30	Start wash-out period
10:45	Welcome in room 2
11:00	Participant working
12:00	Participant diary and EMA
12:15	Start wash-out period and lunch
12:45	Welcome in room 3
13:00	Participant working
14:00	Participant diary and EMA
14:15	Start wash-out period
14:30	Welcome in room 4
14:45	Participant working
15:45	Participant diary and EMA
16:00	Take-off sensors
16:15	Focus group

Appendix B – Pilot study week planning

	Week 1				
	Mon 02-12-24	Tue 03-12-24	Wed 04-12-24	Thu 05-12-24	Fri 06-12-24
Participant 1	p10	p1	p6	p9	p7
Participant 2	p12(no show)		p5	p15	p3
Participant 3			p4		
Participant 4					
	Week 2				
	Mon 09-12-24	Tue 10-12-24	Wed 11-12-24	Thu 12-12-24	Fri 13-12-24
Participant 1				p11	p13
Participant 2					
Participant 3					
Participant 4					

Appendix C – Psychological stress questionnaire

What is your participant number? [ID] - open

What room are you in? [Room] - multipleChoice

First we have a few questions regarding your feelings right now at this moment. We use scales that range from the absence of a feeling to its maximum intensity of the feeling. Please, decide first whether you experience the feeling described by a particular mood word, then to what extent. The scale of answers: 1. I do NOT feel it. 2. I feel it SLIGHTLY, 3. I feel it MODERATELY, I feel it STRONGLY.

Please indicate how AWAKE (wakker) you have felt during this hour.

Please indicate how UNPLEASANT (onprettig) you have felt during this hour.

Please indicate how AT REST (op m'n gemak) you have felt during this hour.

Please indicate how TIRED (moe) you have felt during this hour.

Please indicate how PLEASANT (prettig) you have felt during this hour.

Please indicate how ALERT (alert) you have felt during this hour.

Please indicate how RESTLESS (rusteloos) you have felt during this hour.

Please indicate how POSITIVE (positief) you have felt during this hour.

Please indicate how SLEEPY (slaperig) you have felt during this hour.

Please indicate how NEGATIVE (negatief) you have felt during this hour.

Please indicate how CALM (kalm) you have felt during this hour.

Please indicate how WAKEFUL (oplettend) you have felt during this hour.

Please indicate how GOOD (goed) you have felt during this hour.

Please indicate how JITTERY (onrustig) you have felt during this hour.

Please indicate how RELAXED (ontspannen) you have felt during this hour.

Please indicate how BAD (slecht) you have felt during this hour.

Please indicate how DROWSY (suf) you have felt during this hour.

Appendix D – Sensory diary questions

During this hour in this room, how did it feel working there? [open question]

If you could place 3 positive and 3 negative post-its about the room in the space, where would you place them, and what would you write on them? [open question]

How would you rate the visual environment of this room? [7-point Likert scale]

Please explain your answer [open question]

How would you rate the noise/sounds in your workspace? [7-point Likert scale]

Please explain your answer [open question]

How would you rate the smell in the room? [7-point Likert scale]

Please explain your answer [open question]

How would you rate the touch/feel of the room? [7-point Likert scale]

Please explain your answer [open question]

Which of the senses had the most positive/negative impact on your workday? [multipleChoice]

What was the effect it had? How did this effect come about? [open question]

Appendix E – Semi-structured focus group/interview guide

Introduction

Welcome

As also mentioned in the informed consent, I would like to audio record this conversation, so I can relisten to it when analysing, is that okay?

Introductory questions

I would like to start off by discussing your own experiences with work environments

- Could you please rank the rooms you have stayed in?
- Please explain why you have ranked them in this order

Key questions (Adjusted based on diary data, so modifications will be made, more specific questions and examples will be prompted. Aided by pictures of the rooms as probes) *The green highlighted questions are examples of notes from the diaries, taken with into the focusgroup or interview.*

Transition sentence: Now that we have discussed your general ranking of the rooms, let's dive a little deeper into the separate sensory experiences of smell and touch, and their effect on your stress levels.

- Where they aware of the sensory modality changes?
 - How did they experience the manipulation (smell and touch)
 - Touch is higher in soft environments
 - Soft/no smell smelled musty?
 - Soft/smell was experienced as no smell?
 - Mentioned in the soft environments nice interior, what was nice about it? Why was that nice?
- Ask about their stress levels
 - From the diary, you seemed to feel more pleasant in soft environments? (diary) Can you elaborate on that feeling? Why do you think you felt that way?
 - From the diary, you seemed to get more tired and less alert, but not per se more relax? What is it that you felt? Why do you think that was?
 - What role did your senses play in the feeling this environment created? How did it make you feel emotionally?
 - How did the environment make you feel physically? (e.g. how comfortable was it? Did it irritate you? Was it overstimulating? Did you experience physical reactions?)

How did it play into tacit experiences such as homeliness, professionalism, naturalness?

Closing questions

- How did it feel to pay attention to those sensory experiences in this study? Did it change the experience?

Thanking for participating.

Appendix F - Participant briefing note

Dear Participant,

Thank you for taking part in my thesis research study! Below, you'll find all the important details you need to know about the study, as well as guidelines to follow for your comfort and for the accuracy of the results.

Study Overview

This study aims to explore your **sensory experience** and **psycho-physiological stress** in a working environment. During the study:

- You will work in **four different rooms**, each designed to offer distinct experiences.
- We will measure your **heart rate variability**, **skin conductance**, and **activity** using a wearable sensor and sensors on two fingertips.
- At the end of each hour in a room, you will complete a short questionnaire about your mood and sensory experience.
- At the end of the day, there will be a **focus group discussion** with all participants to reflect on your experiences in greater depth. This starts at 16:15, the location will be made known at the start of the experiment.

Before the Study

To ensure accurate measurements:

- **Avoid excessive alcohol** the evening before the study.
- **Refrain from caffeinated drinks** (e.g., coffee, tea, energy drinks) for at least one hour before the study.
- **Please wear a loose shirt**, as we have to place electrode stickers on your collarbones and underneath your rib cage (please contact Annique if you do not feel comfortable with the placement of these electrodes).
- Download the m-Path app (See logo in the right), via which you can fill in the questionnaires.
- Arrive **on time**, at **Atlas general purpose lab 1: room 9.126**. Please use the toilet before the study, and try to refrain from having to get up during the study.



During the Study

To maintain the quality of the data:

- Perform your **usual solitary tasks** (no meetings or calls).
- **Avoid highly stressful tasks**. If unavoidable, inform the researcher, Annique.
- **Do not adjust the sensors**. If they feel uncomfortable or malfunction, please contact Annique.
- **Do not eat or drink caffeinated items**. Snacks can be enjoyed during wash-out periods. You can bring water or other non-caffeinated drinks.
- Wifi will be available in the rooms
- Take your time to evaluate the room and fill out the questionnaire thoughtfully.
- During wash-out periods, you may walk around or stay in the reserved room. Do make sure to be on time at the next room.

Your Rights

Your participation is voluntary, and you are free to withdraw from the study at any time, without any explanation. Annique will be seated in the hallway of General Purpose lab 1 (atlas 9.126), where you can come to notify me.

We appreciate your cooperation and look forward to learning from your valuable input. If you have any questions or concerns at any point, please don't hesitate to contact Annique.
Thank you for your time and effort!

Sincerely,

Annique Laurens Janse

a.j.laurens.janse@student.tue.nl

Time schedule 1

Time	Activity
09:00	Welcome in room 1
09:15	Participant working
10:15	Participant diary and EMA
10:30	Start wash-out period
10:45	Welcome in room 2
11:00	Participant working
12:00	Participant diary and EMA
12:15	Start wash-out period and lunch
12:45	Welcome in room 3
13:00	Participant working
14:00	Participant diary and EMA
14:15	Start wash-out period
14:30	Welcome in room 4
14:45	Participant working
15:45	Participant diary and EMA
16:00	Take-off sensors
16:15	Focus group

Appendix G – Informed consent form

Information sheet for research project “Sensory Modalities and Stress”

1. Introduction

Eindhoven University of Technology (TU/e) invites you to take part in this research project which aim is to explore how sensory ethnography can incorporate multiple senses and enhance research on psycho-physiological stress in the workplace environment. The gathered data will be used, in an anonymized form, in the master thesis report. You receive this invitation because you have indicated to be willing to participate in this study via the ‘Archie proefpersonenbank’ or personal contact with the researcher. For your participation you will receive a compensation of €18.00 as a sign of our appreciation.

Joining this research project is your choice. Your participation is completely voluntary and does not pose any physical, legal or economic risks. You are not obliged to answer questions you are uncomfortable with, and you can withdraw from the research at any time without explaining why. Declining or withdrawing will not have negative impact for you.

Throughout the research, you might be asked personal questions related to the delicate topic of your perceived stress levels. This data will be reported on completely anonymized. The only people who have insights in the raw data are the involved researchers. During the focus group, other participants are asked not to forward any information heard from other participants, and you are free to answer only what you feel comfortable answering. We ask these questions only for the benefit of the research project.

Before you decide, please read the following information to understand what the project is about, what we expect from you and how we handle your personal data. After reading, you can sign up by completing the attached form.

If you have questions, feel free to contact us (contact details below). You can also discuss this information with people you trust.

2. Who are we?

This research project is conducted by:

Technische Universiteit Eindhoven (TU/e) De Groene Loper 3 5612 AE Eindhoven	Main researcher: Annique Laurens Janse Supervisors: prof.dr.ir. Yvonne de Kort dr.ir. Rianne Appel-Meulenbroek Sophie Schuller
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How to contact us?

Annique Laurens Janse	a.j.laurens.janse@student.tue.nl
Privacy Operations	privacy@tue.nl
Data Protection Officer	dataprotectionofficer@tue.nl

3. What will taking part in the research project involve?

In the research project we will collect your personal data using the following methods:

- Visiting 4 different rooms for an hour so we can ask you about your perceived stress levels and sensory experience in the rooms.
- Measuring your physiological stress by means of electrodermal activity and heartrate variability. The movements are measured by an activity tracker to verify the heart rate irregularities.
- Asking you to fill in various questionnaires during a period of one day, 4 times about your sensory experience of the room, and psychological stress levels.
- A group discussion about the sensory experience in the various rooms, and how it might have effected your emotions, comfort and stress. This discussion will be audio recorded, turned into a transcript.

4. What personal data from you do we gather and process?

We collect and process the following personal data which is necessary for the project purpose:

Category and type	Personal data	Purpose	Retention period
<i>Contact data (regular personal data)</i>	<i>Name, e-mail</i>	<i>To communicate with you and invite you to events</i>	<i>6 months after data collection</i>
<i>Job profile (regular personal data)</i>	<i>Job title</i>	<i>Analyze data concerning your tasks and work environment</i>	<i>6 months after data collection</i>
<i>Student data (regular personal data)</i>	<i>Study name and year</i>	<i>Analyze data concerning your tasks and study-environment</i>	<i>6 months after data collection</i>
<i>Physiological stress data</i>	<i>Electrodermal activity, heartrate variability, movement</i>	<i>To analyse data concerning stress levels in different environments</i>	<i>Permanently</i>
<i>Psychological stress</i>	<i>Self-reported perceived stress</i>	<i>To analyse data concerning stress levels in different environments</i>	<i>Permanently</i>
<i>Sensory perception</i>	<i>Personal perception, experiences, memories concerning the environments</i>	<i>To analyse the experience of an environment through certain sensory modalities</i>	<i>Permanently</i>

Your data is retained only for the time period as specified in the table. Keeping your data for this period helps us to comply with scientific principles, such as producibility and verification.

After this period, your personal data will be deleted or anonymized to ensure it can no longer be linked to you. Unless you explicitly agree to the use of your identifiable information in publications (for example your quotes or your name).

5. Stopping your participation and your rights

If you end your participation in the research we will not use your data anymore from that moment on.

For questions, ending your participation or complaints, please contact the researcher via the contact details as provided in the table under section 2.

For concerns or questions about the handling of personal data, you can e-mail the data protection officer as indicated in the table under section 2. You can also file a complaint with the Dutch data protection authority: the Autoriteit Persoonsgegevens.

You have the right to request access, rectification, objection, erasure or adaptation of your data. Submit your request through team Privacy Operations via the contact details as provided in the table under section 2.

6. Legal basis for processing your personal data

The TU/e processes your personal data to conduct scientific research, which is the university's public task as stated in article 1.3 of the Dutch Wet Hoger onderwijs en Wetenschappelijk onderzoek. The TU/e always follows the applicable codes of conduct for research integrity and the scientific standards when conducting research.

7. Who has access to your personal data?

Only authorized employees involved in the research have access to your personal data if this is necessary for their tasks. The authorized employees will keep your personal data confidential.

Other parties/processors that have access to the data are listed in the table below:

Party/processor	Why access?	Category	Processing within the European Economic Area?
Microsoft Onedrive (Netherlands)	Storage solution		Yes
m-Path	Survey tool		Yes
maxQDA	Data analysis tool		Yes
Samsung	Data collection device		Yes
R	Data analysis tool		yes

TU/e has a suitable agreement with these parties to protect your personal data. We will not share your personal data with any other party, unless we are required to do so by law.

TU/e will process your personal data within the European Economic Area (EEA) by storing your data on a server inside the EEA. In addition, TU/e has implemented appropriate technical and organizational measures to protect your personal data. These measures include using centrally managed and verified research- and storage tools.

8. Future research

During the research project, your personal data may also prove useful for new, socially important research. In such case, we would like to reuse your data for the new research. This will only happen if (1) your personal data is truly necessary, (2) the recognized ethical standards for scientific research are followed, and (3) the new research objectives align with the current research objectives. If your personal data is used in future research, we will take all reasonable steps to inform you about this. You can object to the use of your personal data for new research.

We might use anonymized data for new purposes such as research or education. We will ensure the data cannot be linked to you and we will not disclose anything that makes you identifiable.

This research has been assessed and approved by the ethical committee of Eindhoven University of Technology.

Consent form for participation by an adult

By signing this form, I confirm:

1. I have enough information about the research project from the separate information sheet. I have read it and I had the chance to ask questions, which have been answered to my satisfaction.
2. I take part in this research project voluntarily. There is no explicit or implicit pressure for me to take part in this research project and I understand I can stop my participation at any moment, without explaining why. I do not have to answer any question I do not want to answer.
3. I know my personal data will be collected and used for the research, as explained to me in the information sheet.

Furthermore, I consent to the following parts of the research project:

4. I consent to my personal data, such as audio taken of me, answers or quotes I gave during the research, to be used by the researcher in publications and/or in presentations – without including my name.

YES ☐ NO ☐

Name of Participant:

Signature:

Date:

Name of researcher:

Signature:

Date:

Appendix H – Thematic coding scheme

Code System	Transc...	Transc...	Transc...	Transc...	Transc...	transcr...	transcr...	SUM
Method								9
Stimulating								3
low stimulation, no distraction								1
understimulation/bored								14
overstimulation								10
Positively stimulating								14
Vision								24
Hearing								4
mental state								8
tired/dull								8
alert/awake								7
positive feeling								25
negative/annoyed								18
calm/relaxed								12
stress								3
distracted								8
negative distraction								22
positive distraction								6
focused								13
Touch								11
negative touch								15
Soft								25
temperature								5
visually embodied touch								5
Unnoticed touch								2
Smell								16
negative smell								15
positive smell								20
unnoticed								14
Environment associations								13
Social setting								6
office								10
homely								32
Σ SUM	20	56	94	68	82	40	38	398

